Tools Integration with n8n

Class three

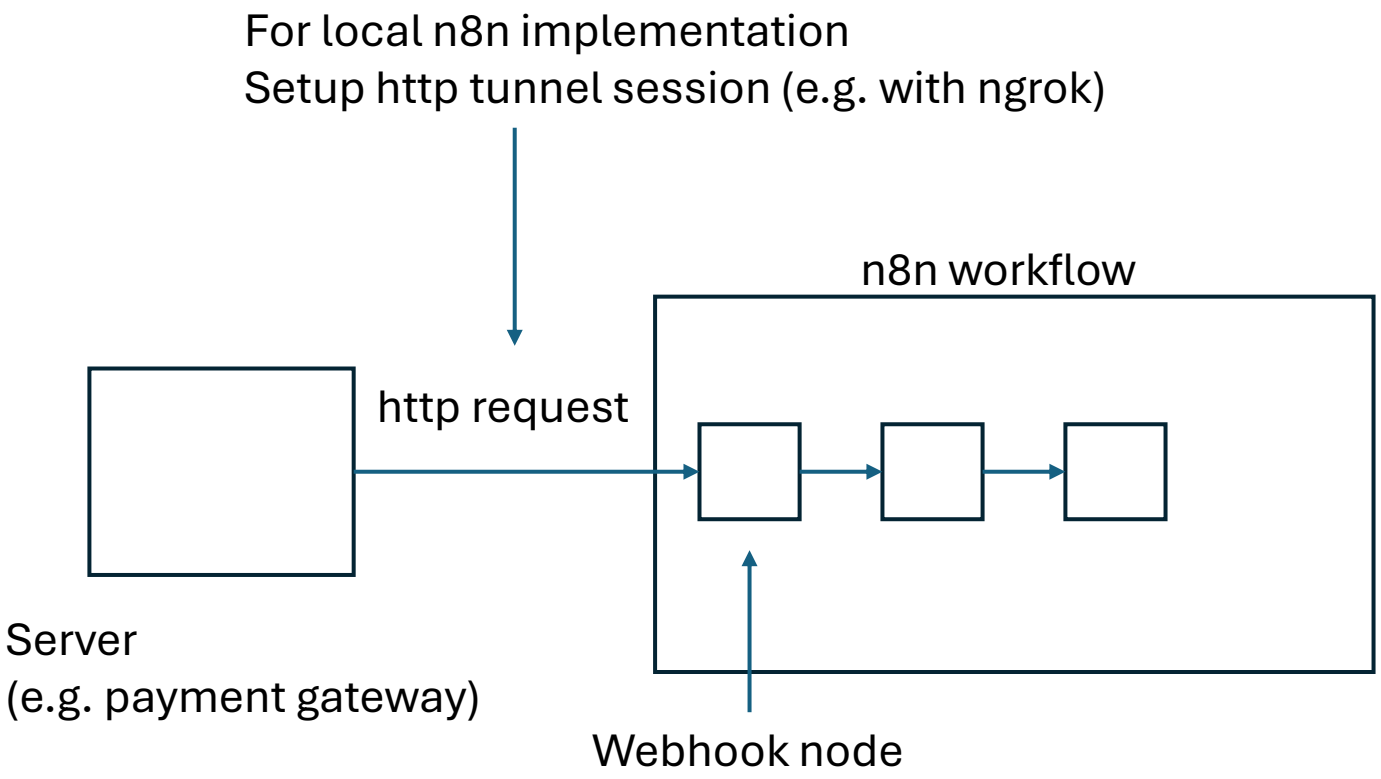
Agenda

- Web tunnelling and Web hooking
- AI chatbot
- Setup Google service API
- Web content summarization with LLM



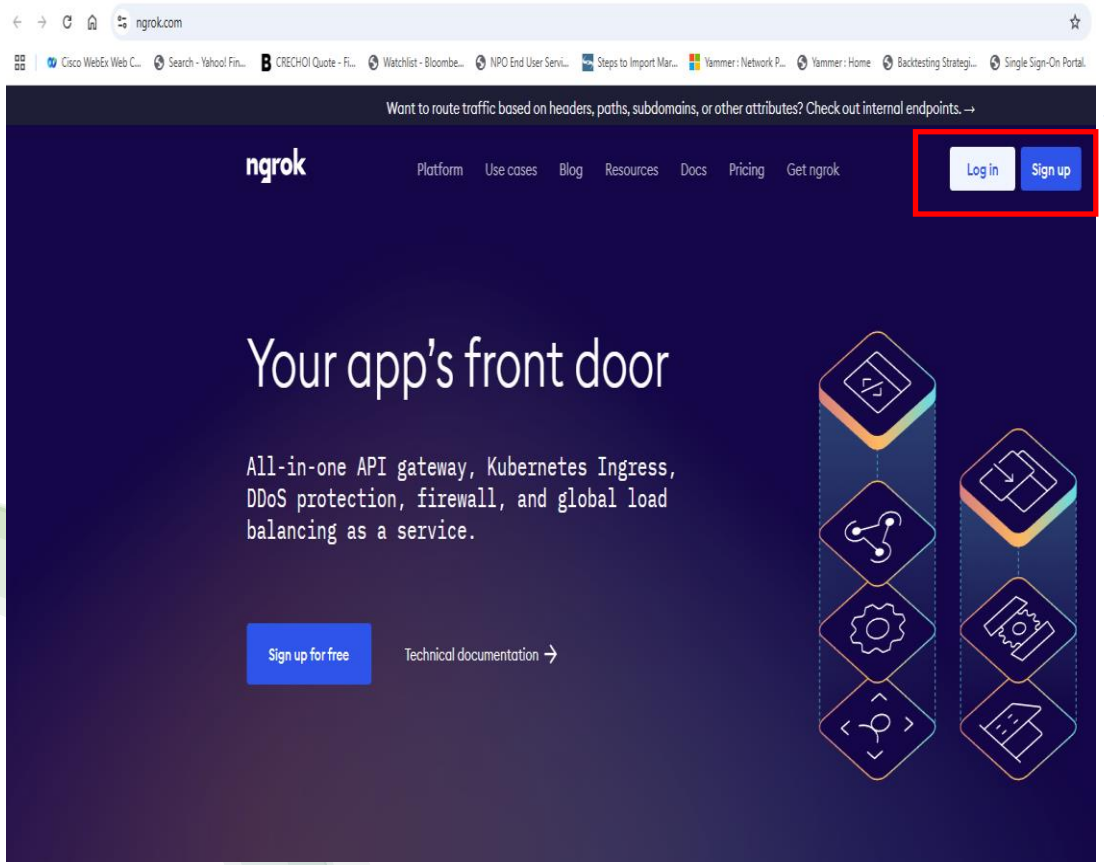
Setup of Web Tunneling and Web Hooking

Webhook


For local n8n implementation
Setup http tunnel session (e.g. with ngrok)
n8n workflow
http request
Server
(e.g. payment gateway)
Webhook node

Building secure tunnel with Ngrok

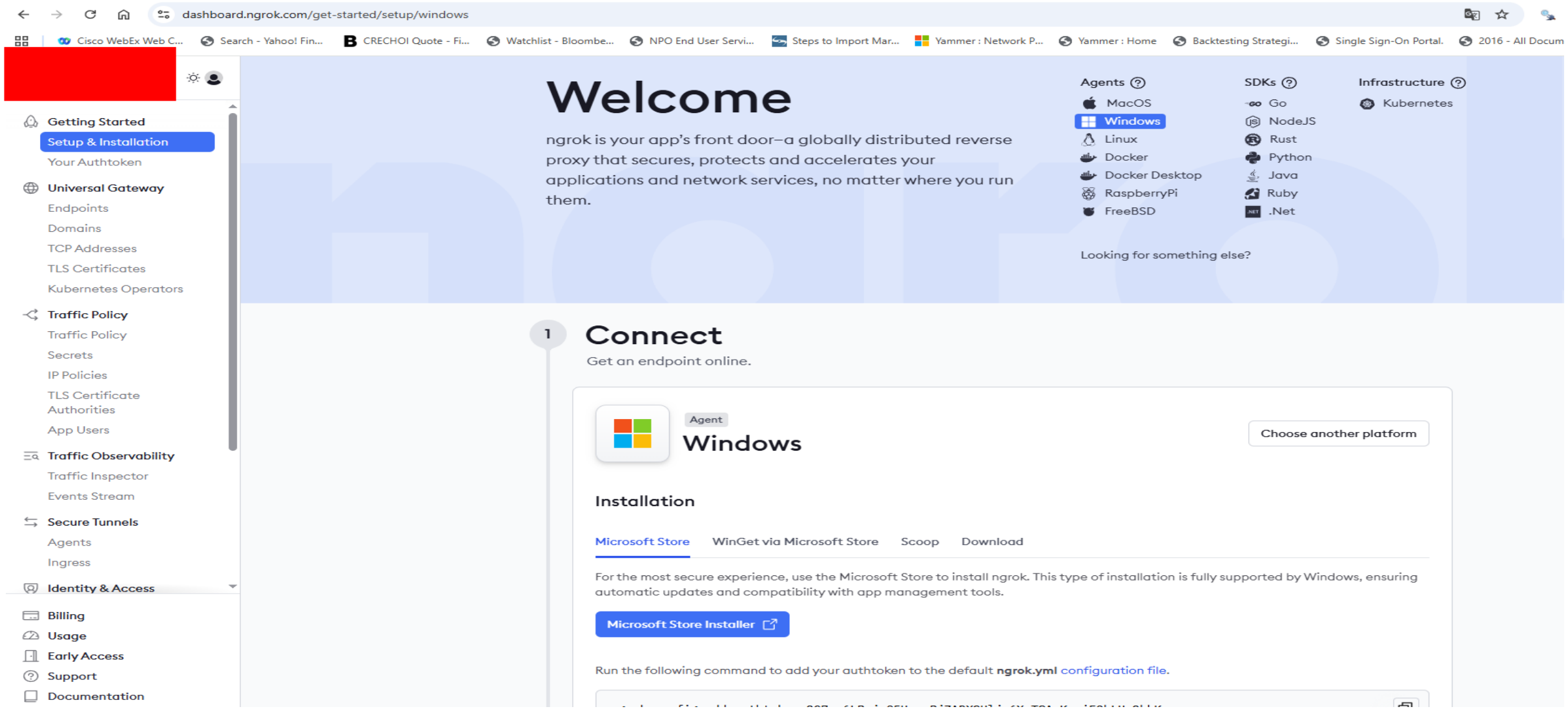
Login or Sign Up



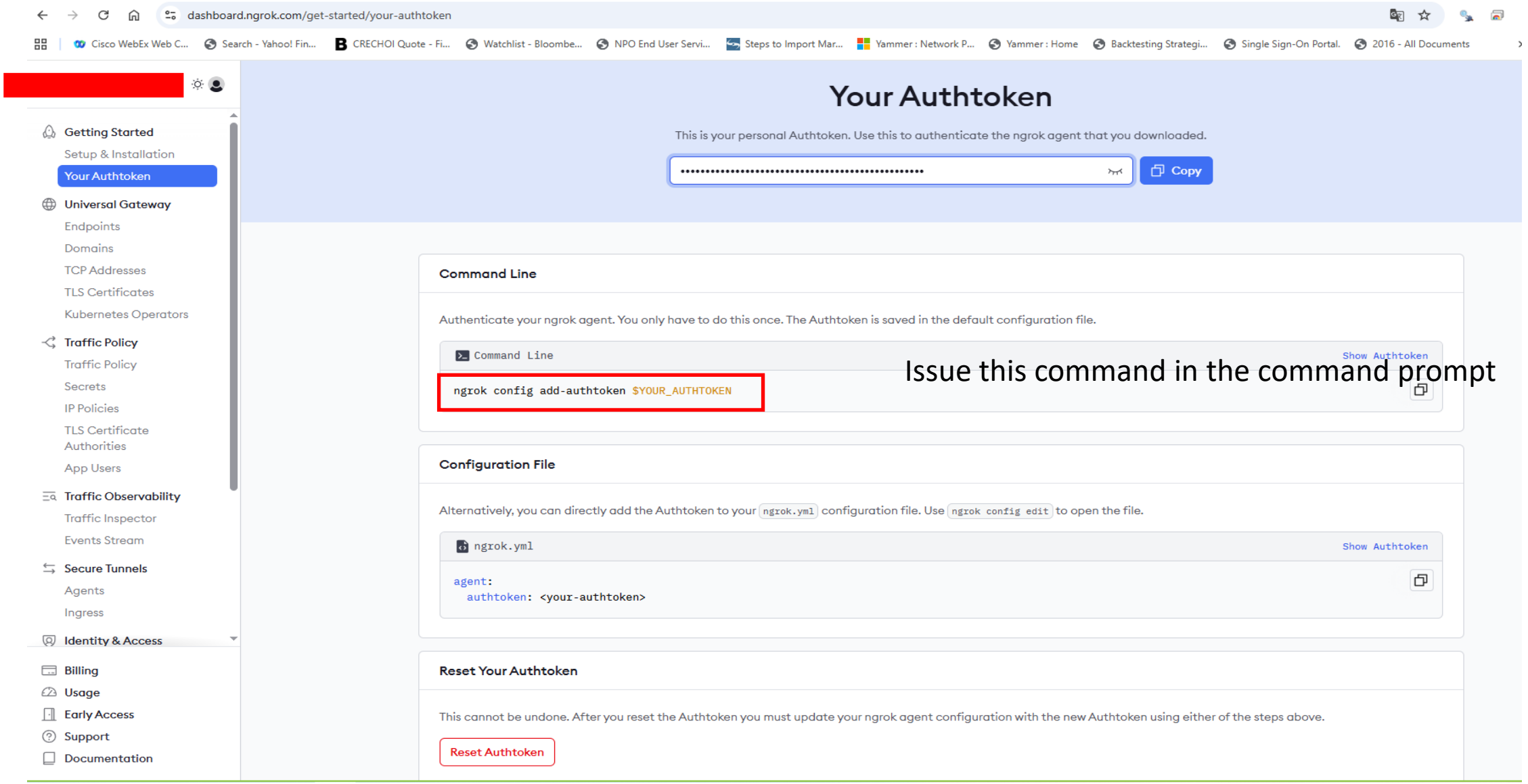
Ngrok Overview:

- **Tool Purpose**: Ngrok creates a secure tunnel from the public internet to a local server on your private network.
- **Key Benefits and Functions:**
- **Tunneling**:
  - Exposes local web servers to the internet through secure tunnels.
  - Allows sharing of a web URL globally that directs to your local application.
- **Secure Connections**:
  - Encrypts all traffic through ngrok tunnels.
  - Protects data from eavesdropping during transmission.
- **Simplifies Configuration**:
  - Removes the need for a public IP address.
  - Eliminates the requirement for port forwarding on routers.

Installation of ngrok in local machine



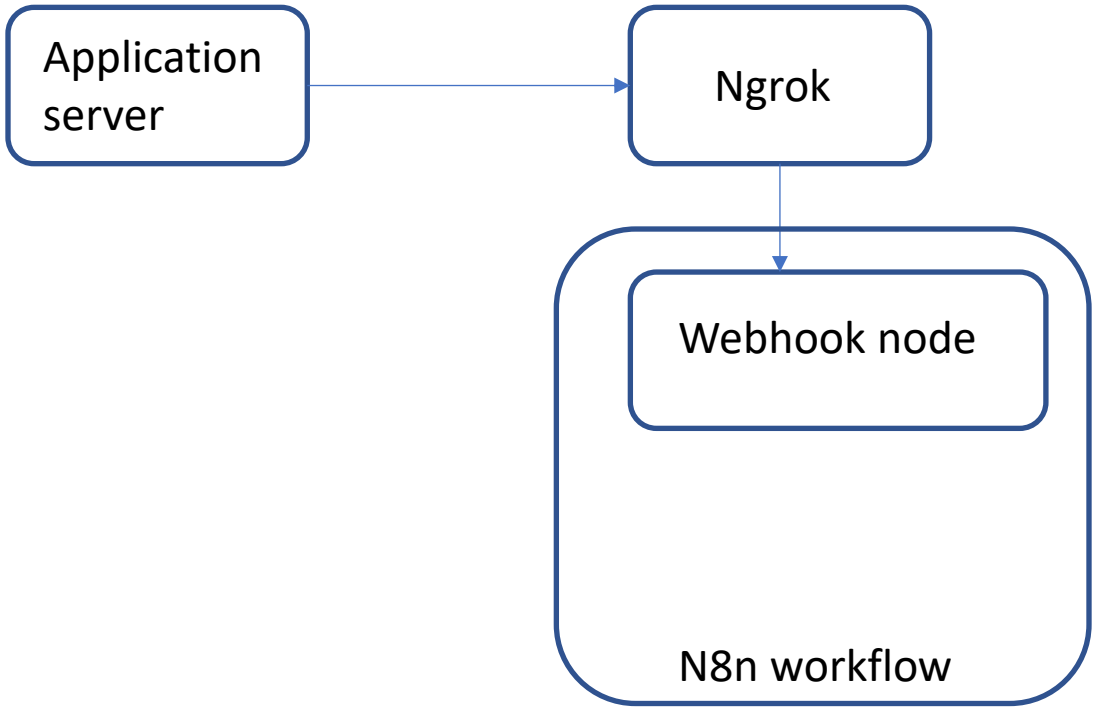
Configure the token in the local machine



Setup Webhooking with Ngrok and Python

Domain name

URL: https://abcdexxxx



Token (get it from Ngrok web site)

1 Configure the token for ngrok session

```
D:\ngrok>

D:\ngrok>ngrok config add-authtoken
```

2 Setup remote tunnel session under local host

```
D:\ngrok>

D:\ngrok>ngrok http http://localhost:5000
```

Port number used for forwarding in the tunnel

3 Copy the URL link assigned by Ngrok

```
Command Prompt - ngrok  http http://localhost:5000
ngrok

   ngrok? We're hiring https://ngrok.com/careers

Session Status                online
Account                       vickslee@gmail.com (Plan: Free)
Update                        update available (version 3.18.4, Ctrl-U to update)
Version                       3.2.2
Region                        Asia Pacific (ap)
Latency                       54ms
Web Interface                 http://127.0.0.1:4040
Forwarding                    https                                -> http://localhost:5000

Connections                   ttl     opn     rt1     rt5     p50     p90
                              2       0       0.00    0.00    0.33    0.33

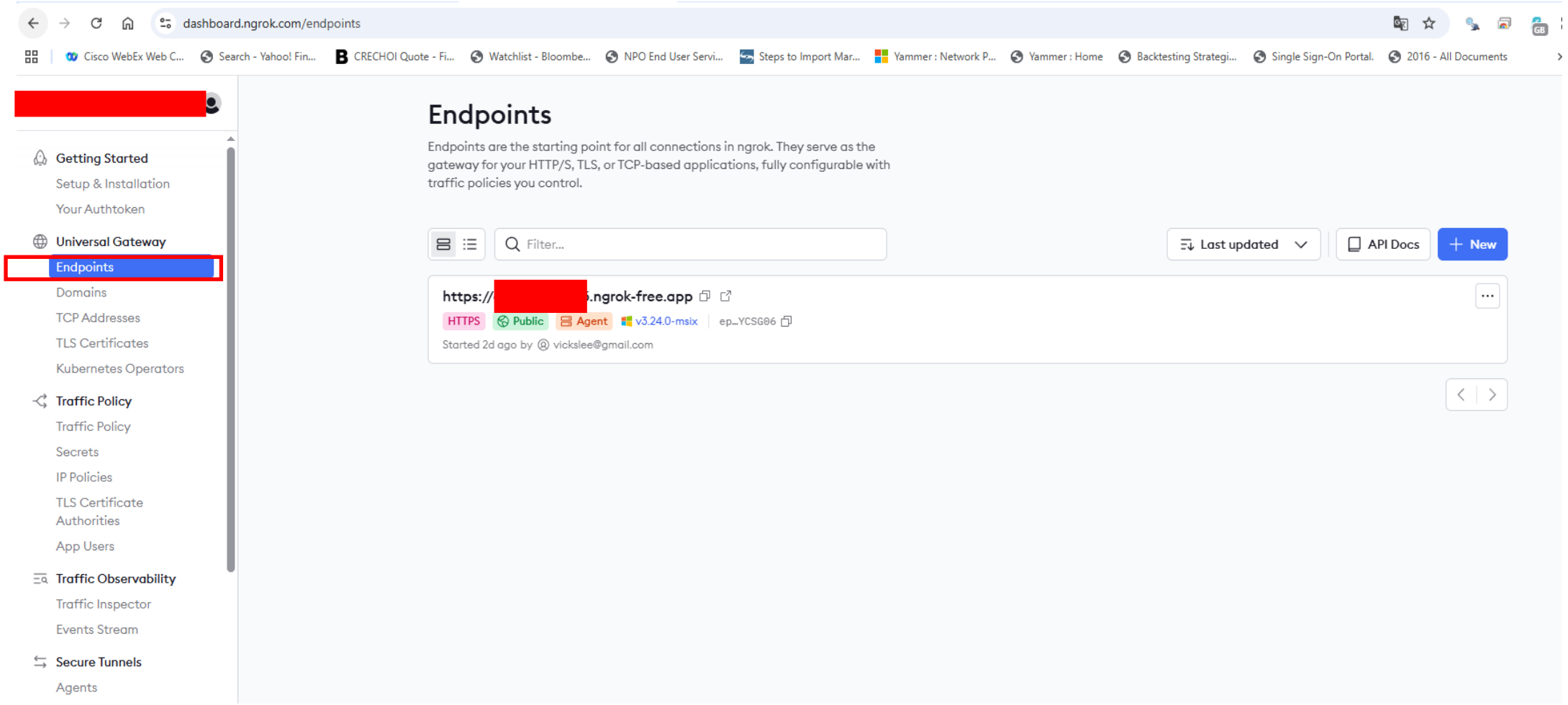
HTTP Requests
-------------

POST /webhook                 200 OK
POST /webhook                 200 OK
```

Web Url assigned by Ngrok

Remark: adjust the port number for n8n (from 5000 to 5678)

Check the endpoint in ngrok



Setup Webhook in local server
n8n setup

- Use the following command **under power shell** in windows and setup the webhook mapping in your local server

$env:WEBHOOK_URL = "https://YOUR-NGROK-SUBDOMAIN.ngrok.io/"

- Use the following command if you want to setup a permanent mapping

setx WEBHOOK_URL " https://YOUR-NGROK-SUBDOMAIN.ngrok.io/"

Use the following command if you want to check the current setting:

echo $env:WEBHOOK_URL

Setup Webhook in local server
Checking the webhook details in Telegram

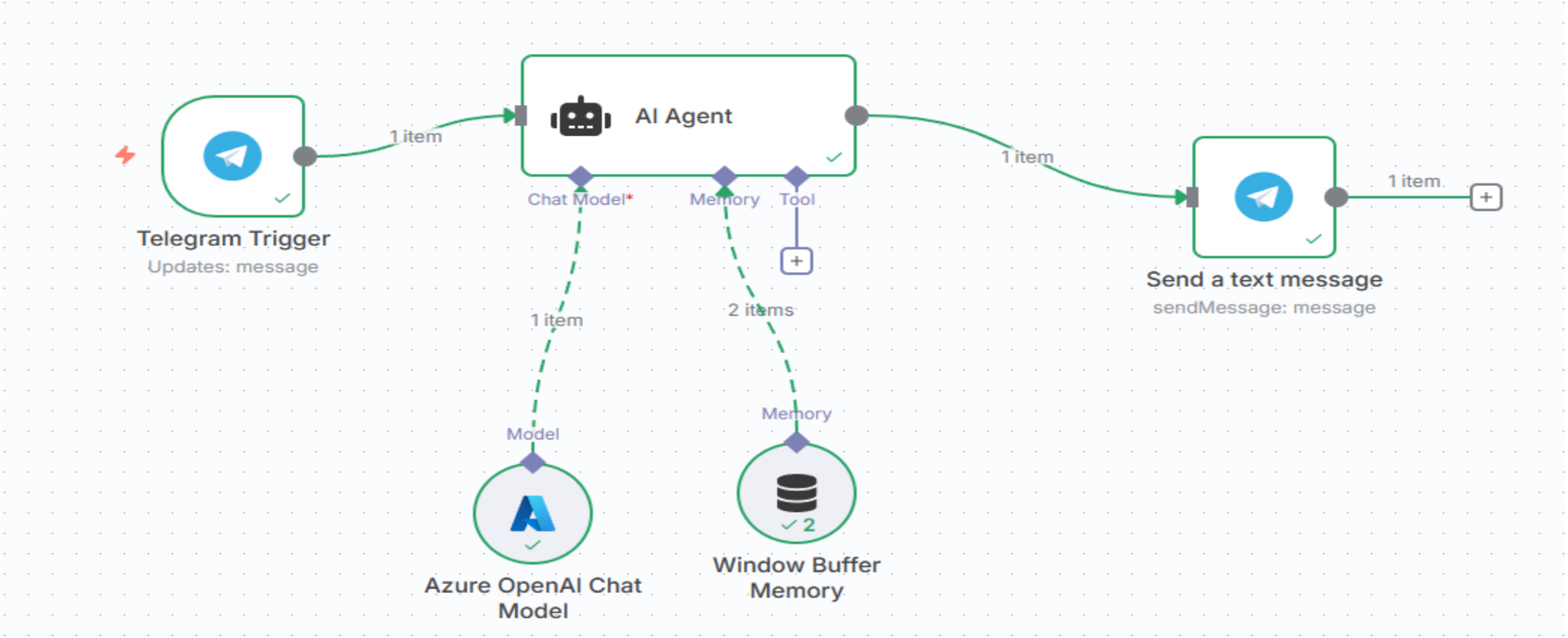
- Use the following link to check the webhook details in telegram

https://api.telegram.org/bot<YOUR_BOT_TOKEN>/getWebhookInfo

{"ok":true,"result":{"url":"https://[redacted]/webhook","has_custom_certificate":false,'

AI Chatbot

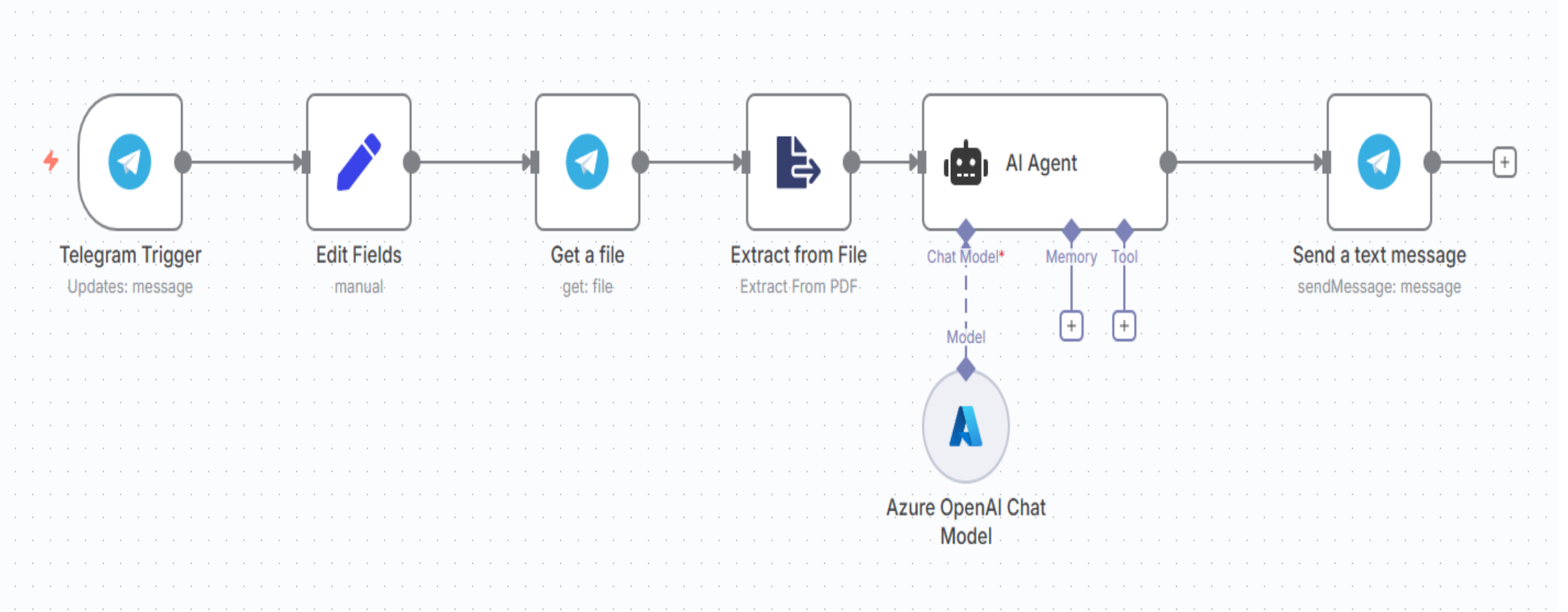
Sample Workflow with n8n



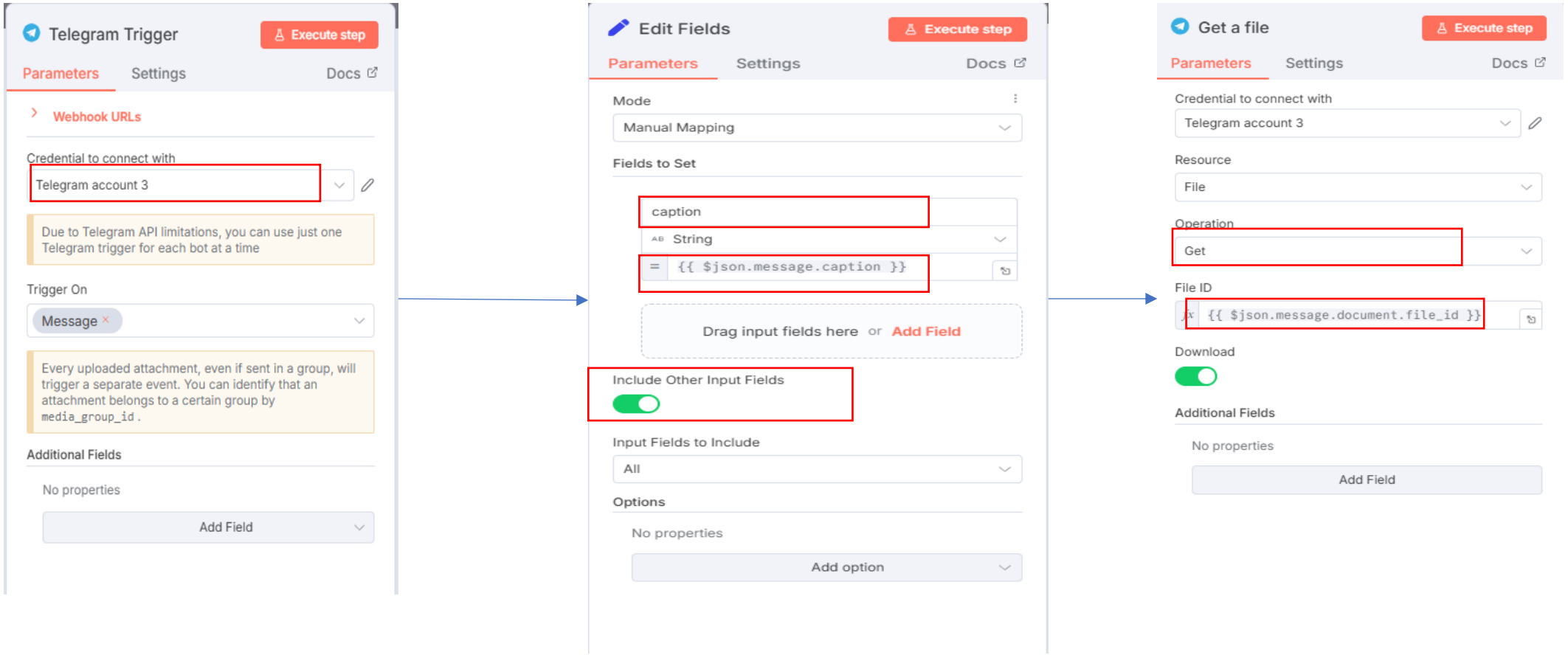
Exercise

AI Chatbot with attached pdf file

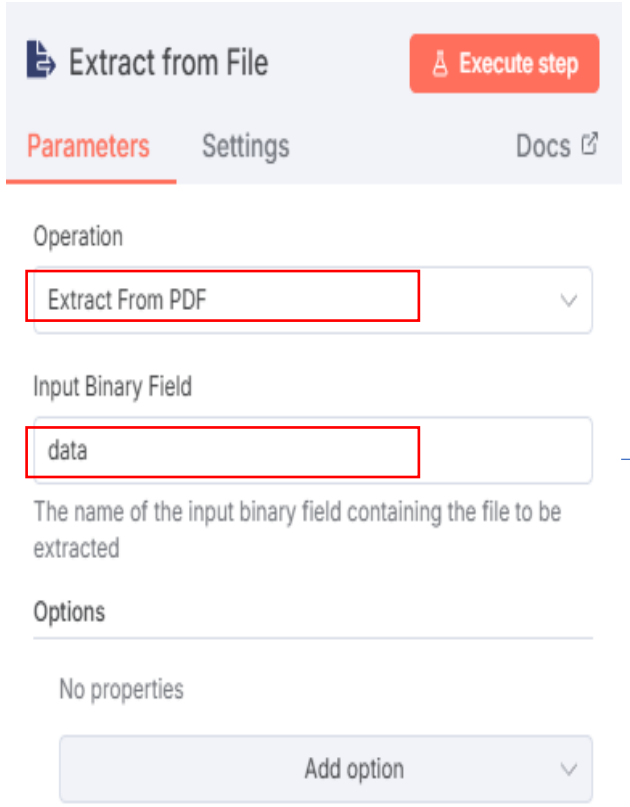
AI Chatbot with attached pdf file

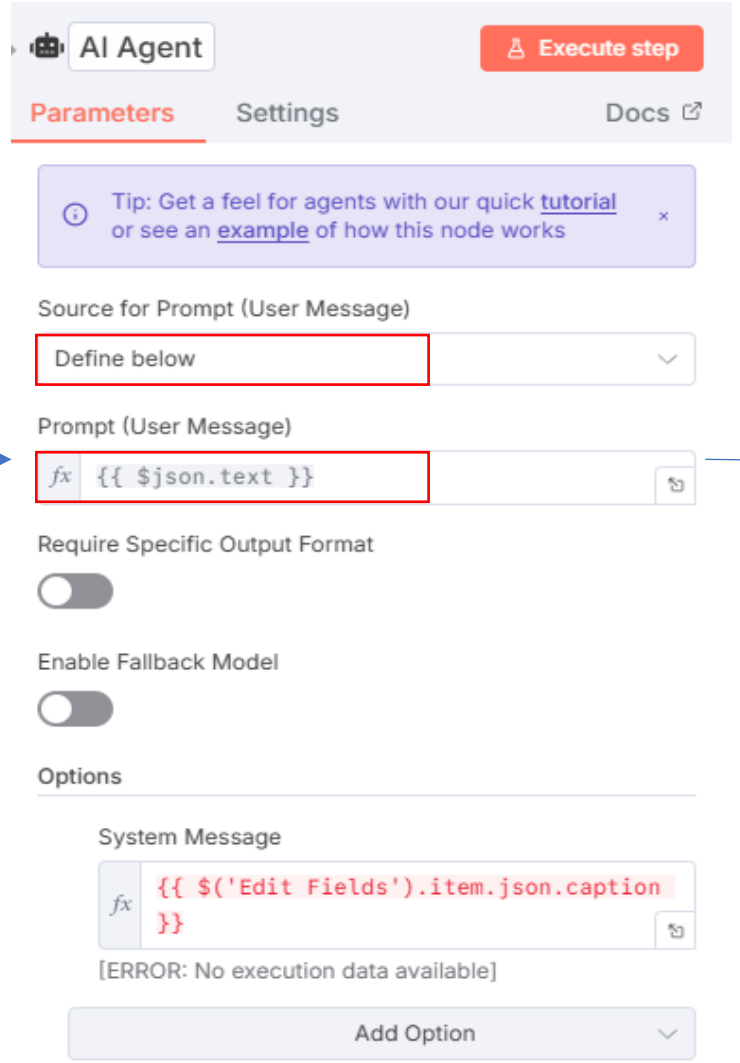


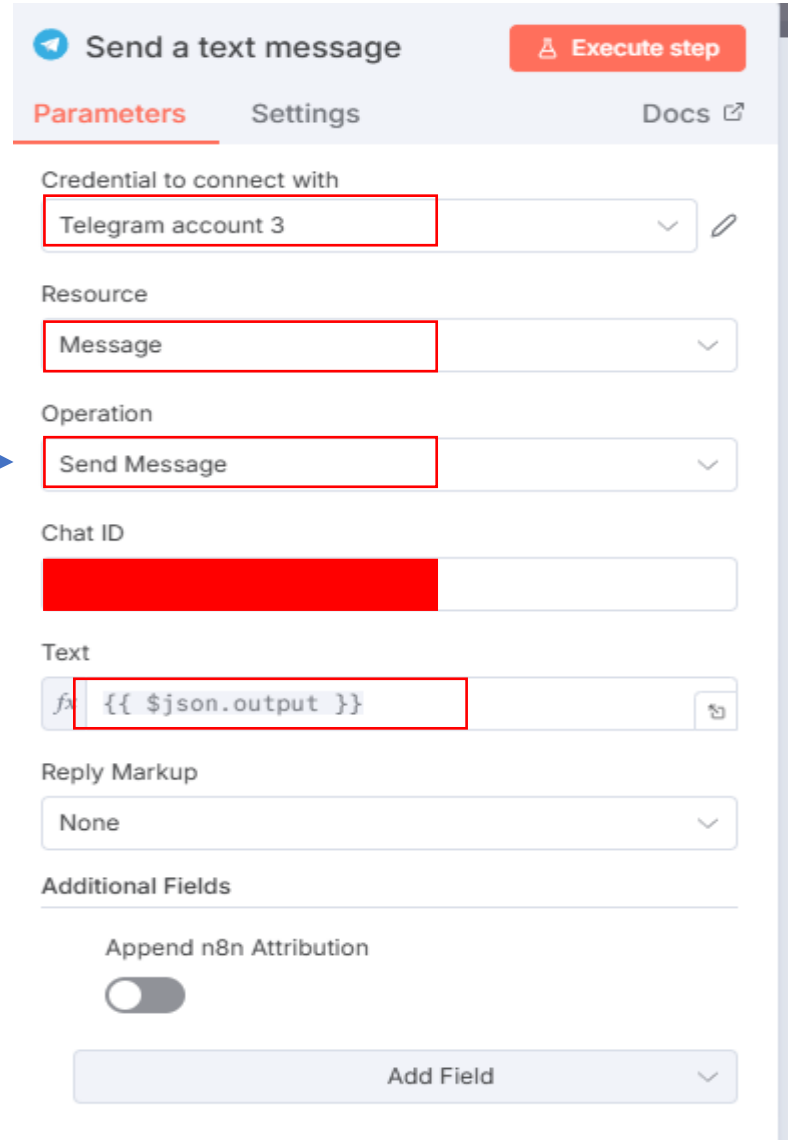
Steps-1/2


Telegram Trigger
Execute step
Parameters
Settings
Docs
Webhook URLs
Credential to connect with
Telegram account 3
Due to Telegram API limitations, you can use just one Telegram trigger for each bot at a time
Trigger On
Message
Every uploaded attachment, even if sent in a group, will trigger a separate event. You can identify that an attachment belongs to a certain group by media_group_id .
Additional Fields
No properties
Add Field
Edit Fields
Execute step
Parameters
Settings
Docs
Mode
Manual Mapping
Fields to Set
caption
String
{{ $json.message.caption }}
Drag input fields here or Add Field
Include Other Input Fields
Input Fields to Include
All
Options
No properties
Add option
Get a file
Execute step
Parameters
Settings
Docs
Credential to connect with
Telegram account 3
Resource
File
Operation
Get
File ID
{{ $json.message.document.file_id }}
Download
Additional Fields
No properties
Add Field

Steps-2/2


Extract from File
Execute step
Parameters
Settings
Docs
Operation
Extract From PDF
Input Binary Field
data
The name of the input binary field containing the file to be extracted
Options
No properties
Add option


AI Agent
Execute step
Parameters
Settings
Docs
Tip: Get a feel for agents with our quick tutorial or see an example of how this node works
Source for Prompt (User Message)
Define below
Prompt (User Message)
{{ $json.text }}
Require Specific Output Format
Enable Fallback Model
Options
System Message
{{ $('Edit Fields').item.json.caption }}
[ERROR: No execution data available]
Add Option


Send a text message
Execute step
Parameters
Settings
Docs
Credential to connect with
Telegram account 3
Resource
Message
Operation
Send Message
Chat ID
Text
{{ $json.output }}
Reply Markup
None
Additional Fields
Append n8n Attribution
Add Field

Setup Google API

Types of Google API

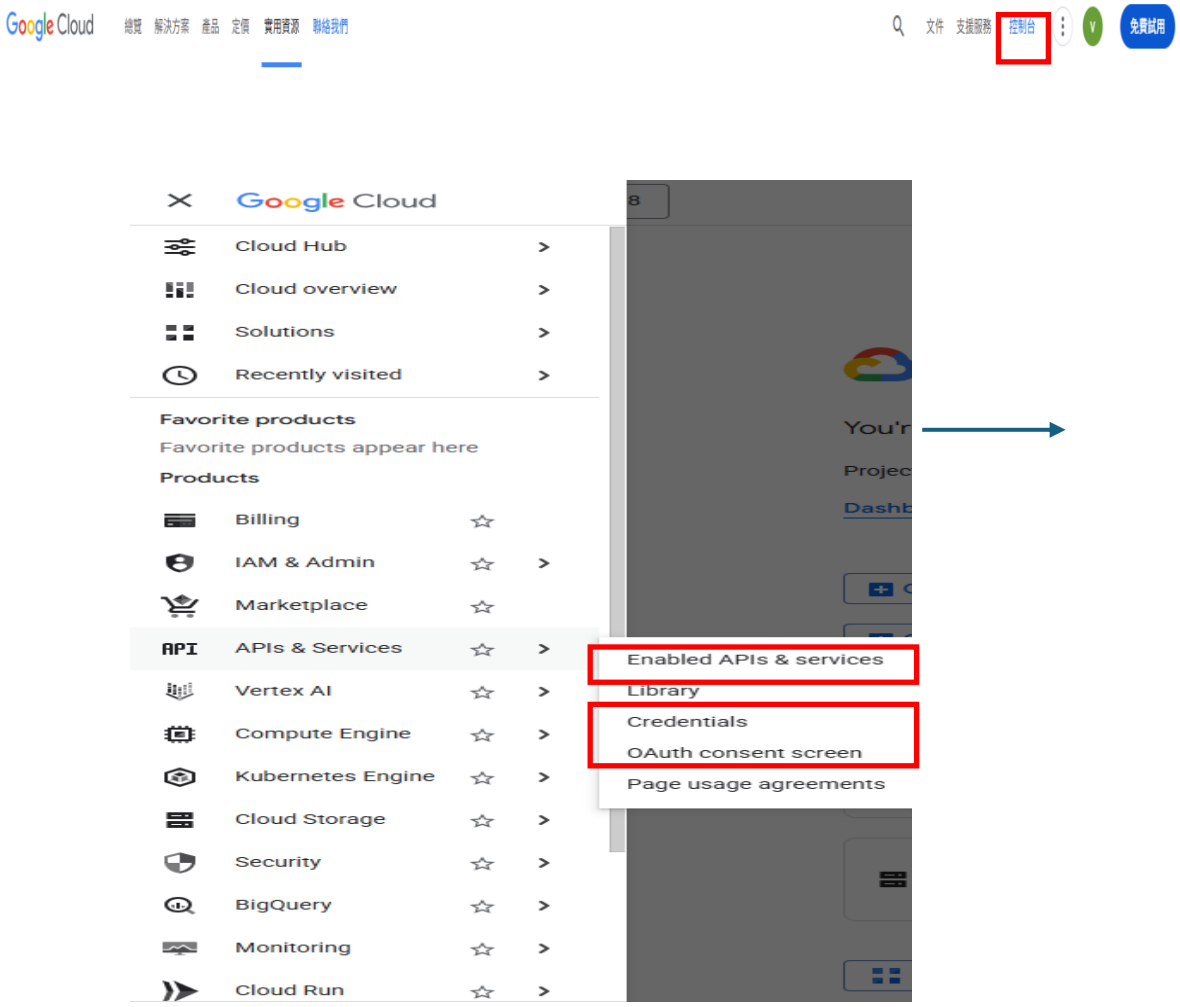
🚦 Quick comparison

Feature	OAuth (User-based)	Service Account (Robot-based)
Represents	A human user account	A non-human account (bot)
Consent needed	Yes (user clicks "Allow")	No (uses key)
Works with "My Drive"	✅ Yes	❌ No (unless shared)
Works with Gmail	✅ Yes	❌ No (unless domain delegation)
Token lifetime	Needs refresh, stable in Production mode	Stable, key-based
Best for	Personal assistants, Gmail, Calendar, Drive automation	Server-to-server, backend jobs, shared drive processing

Need to share the Resource (e.g. google sheet) with the bot

Setup of OAuth (User Based)

Overall Steps (OAuth)



Create new project

Specify the required services

Gmail/Google Drive etc

Configure the OAuth Consent Screen

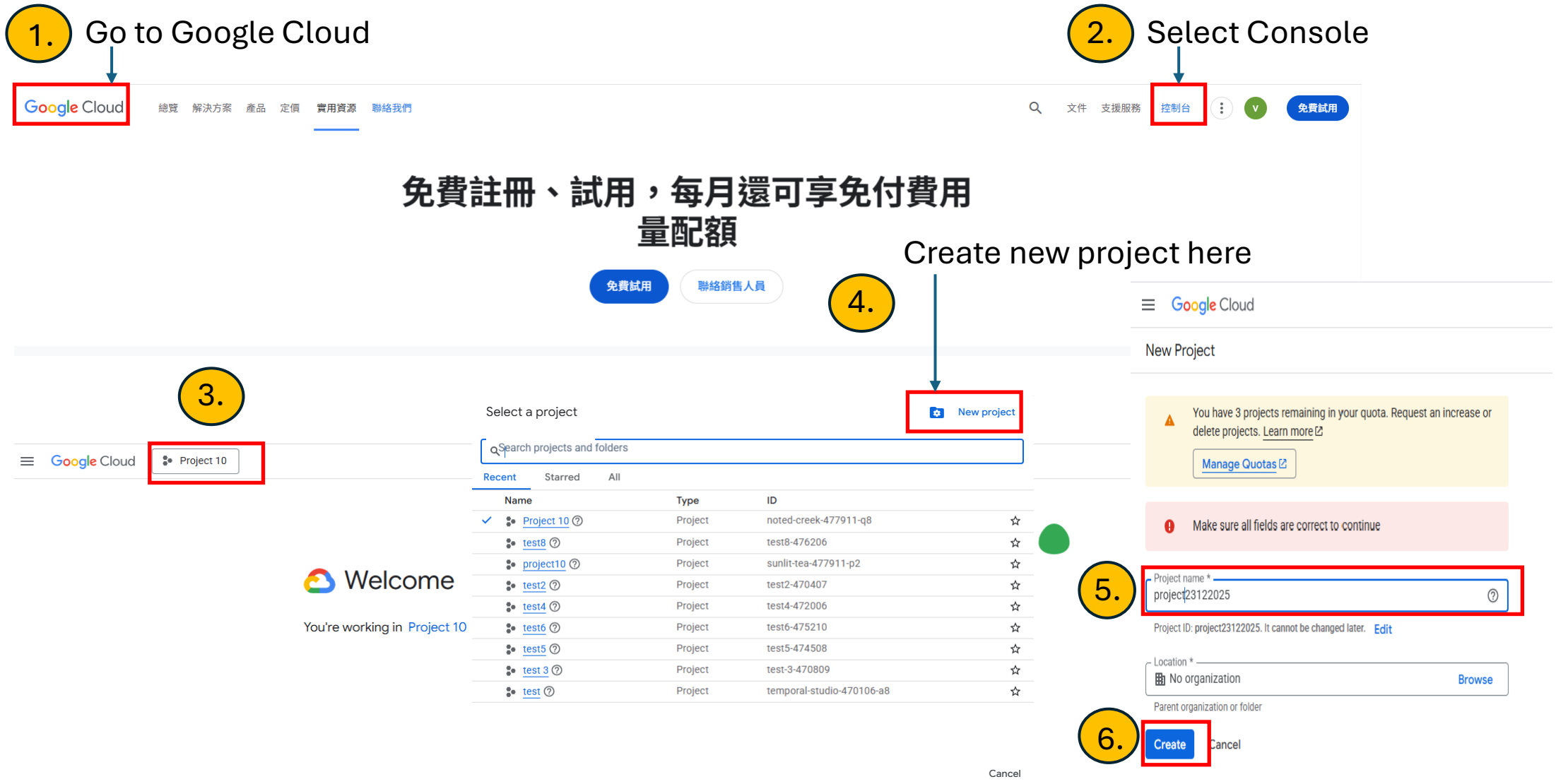
URL callback from Dify/n8n

Configure the OAuth Client

Dify/n8n

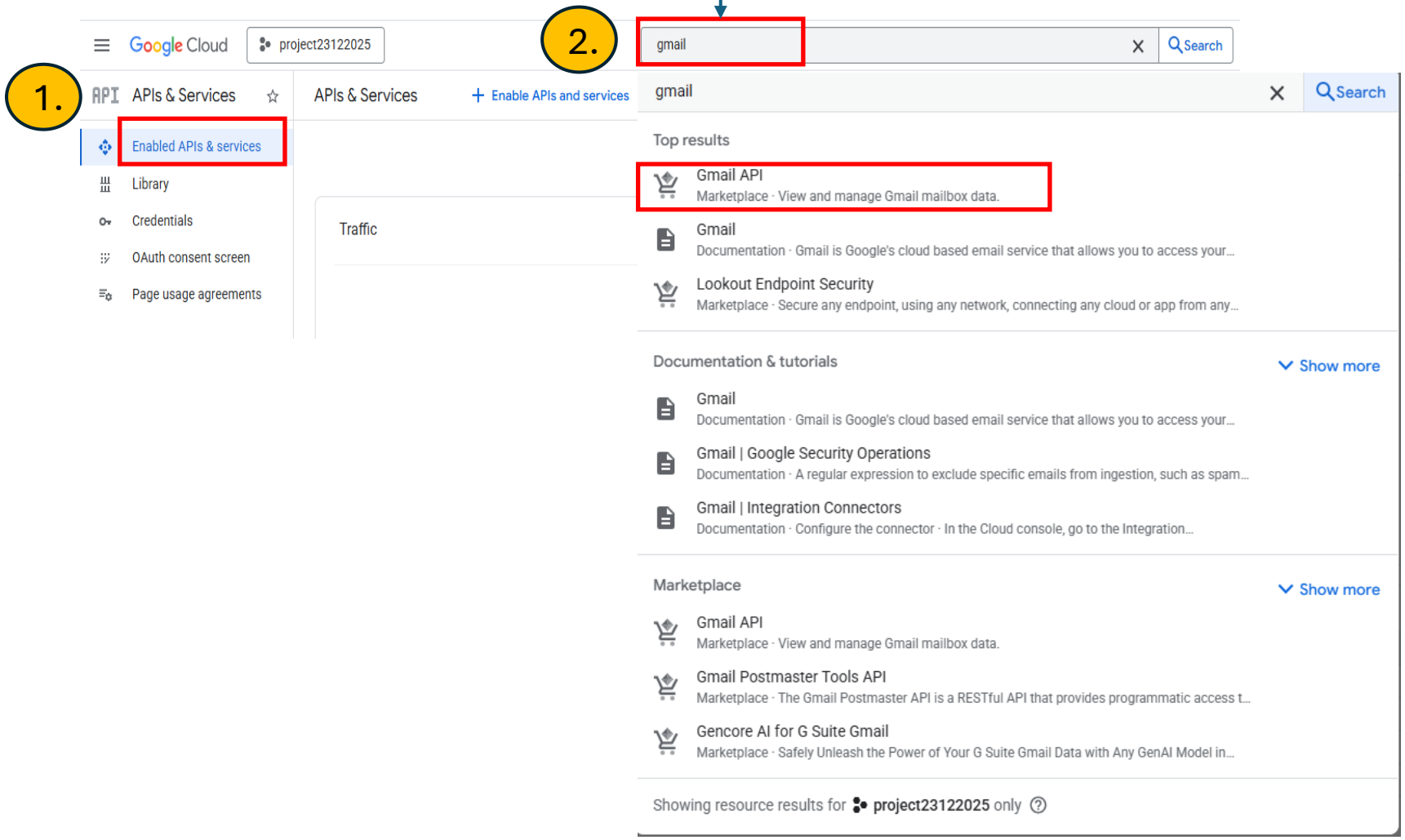
Fill in other information (test users)

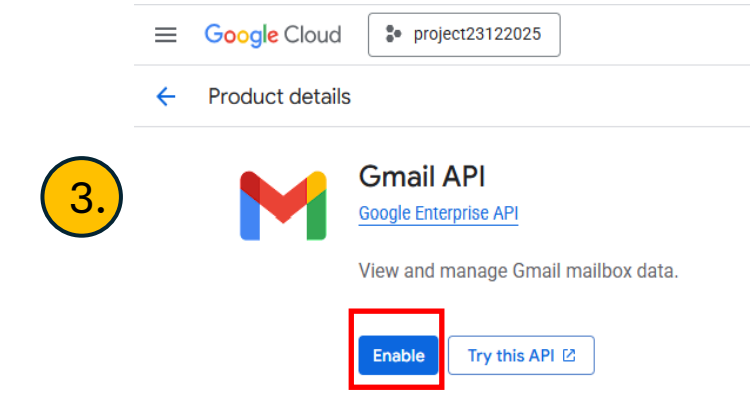
Create New Project



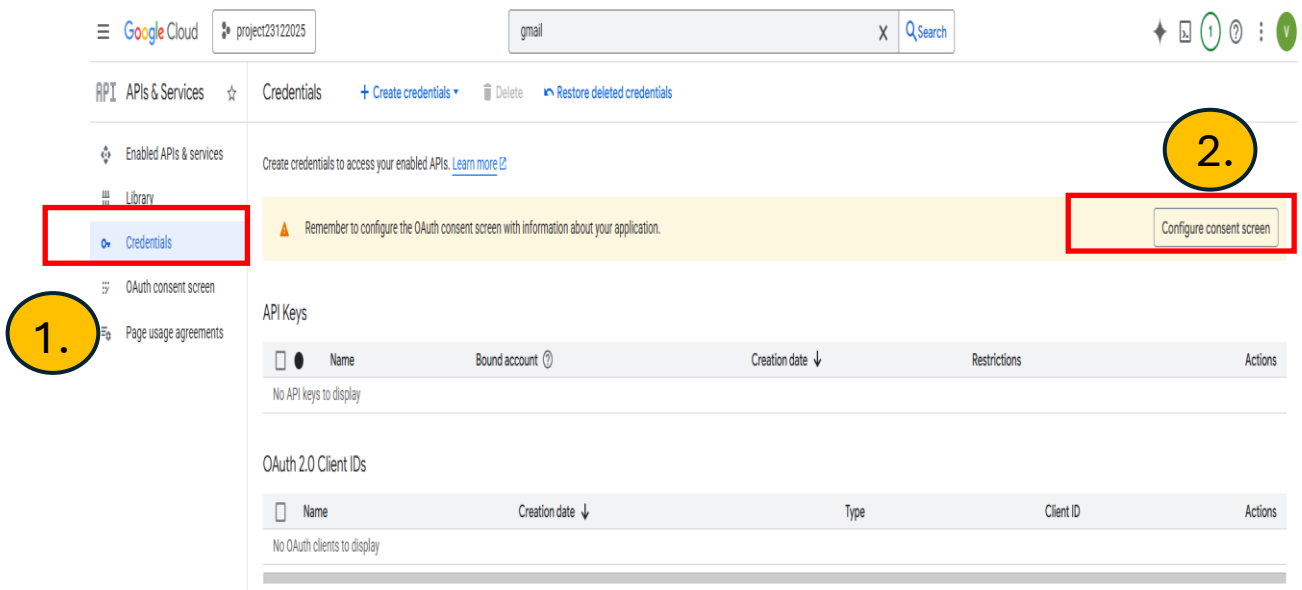
Enable API & Services

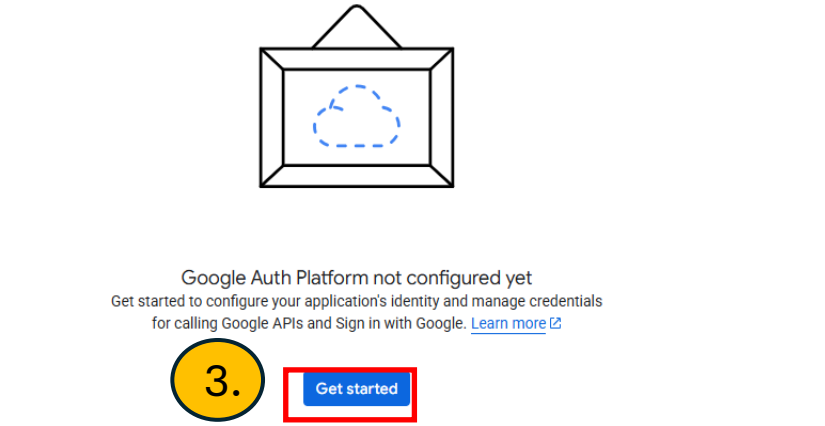
Example: Enable Gmail service

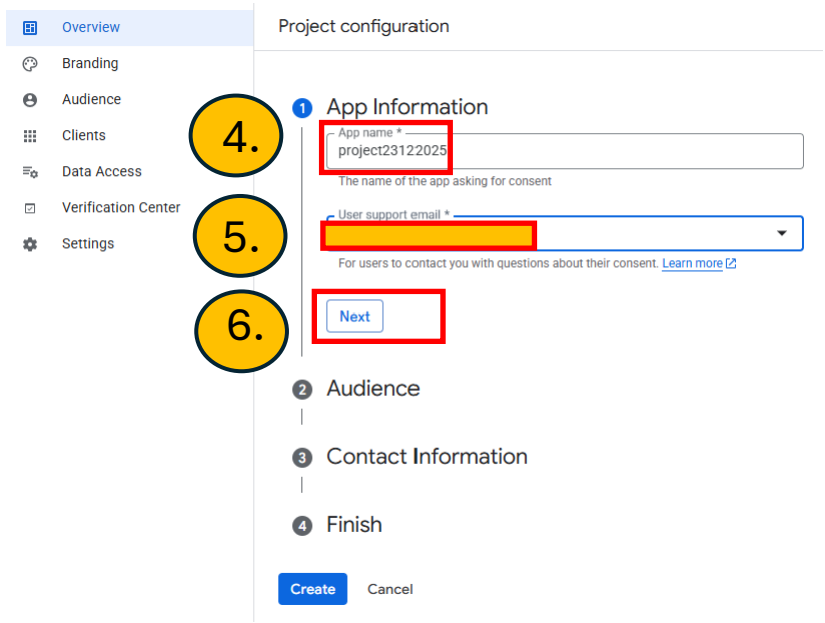




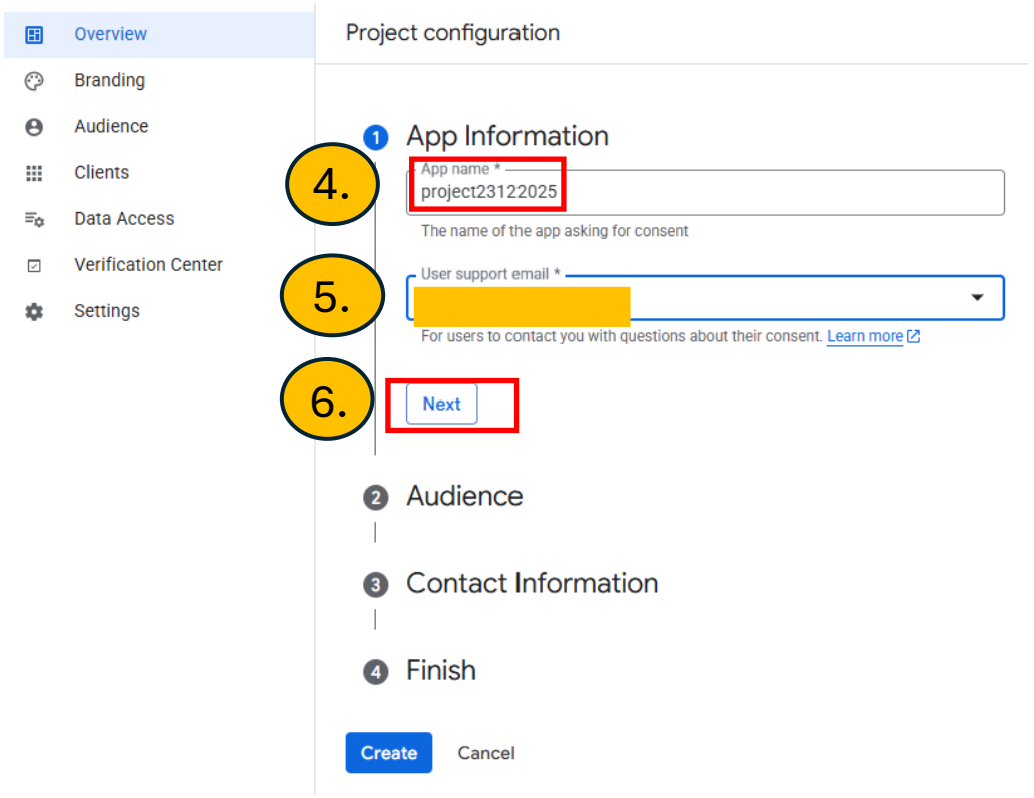
Setup Consent Screen-1/3

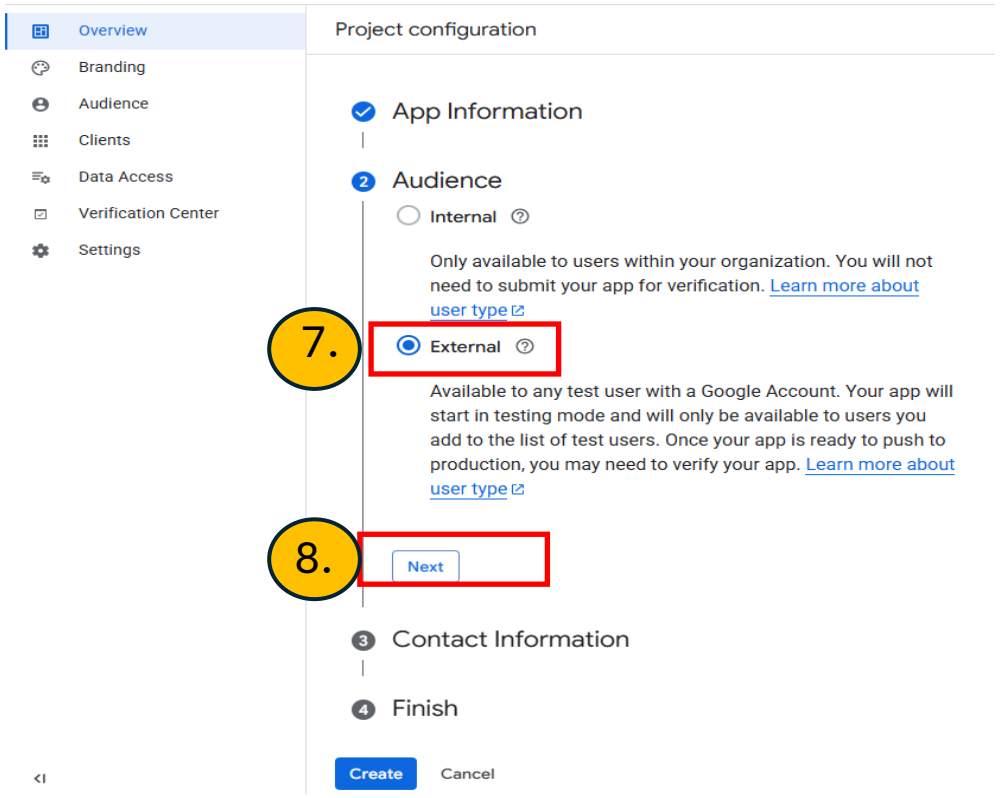




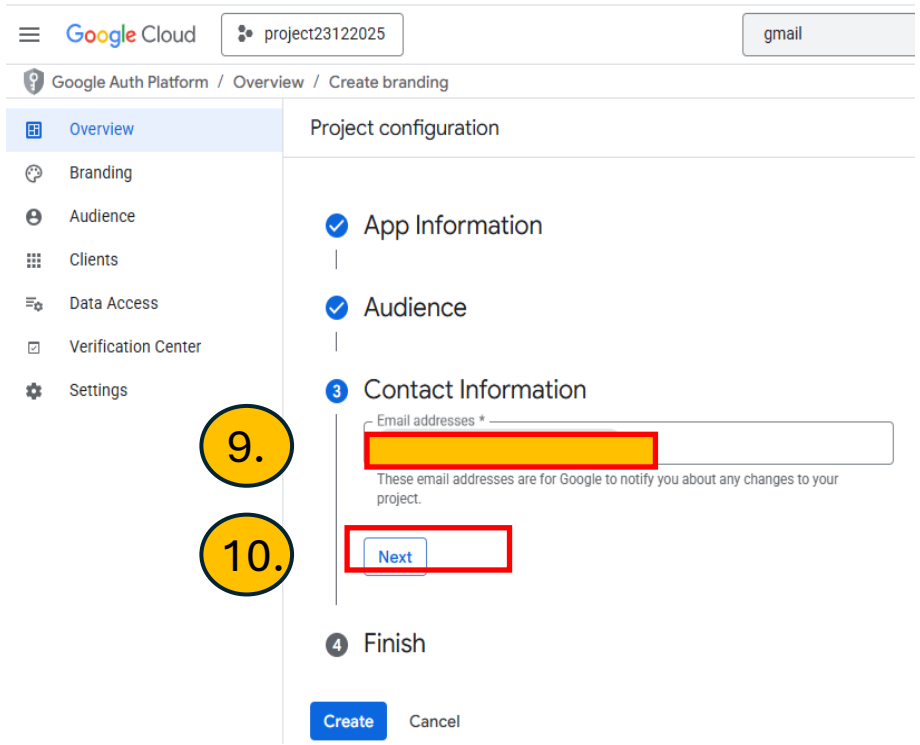


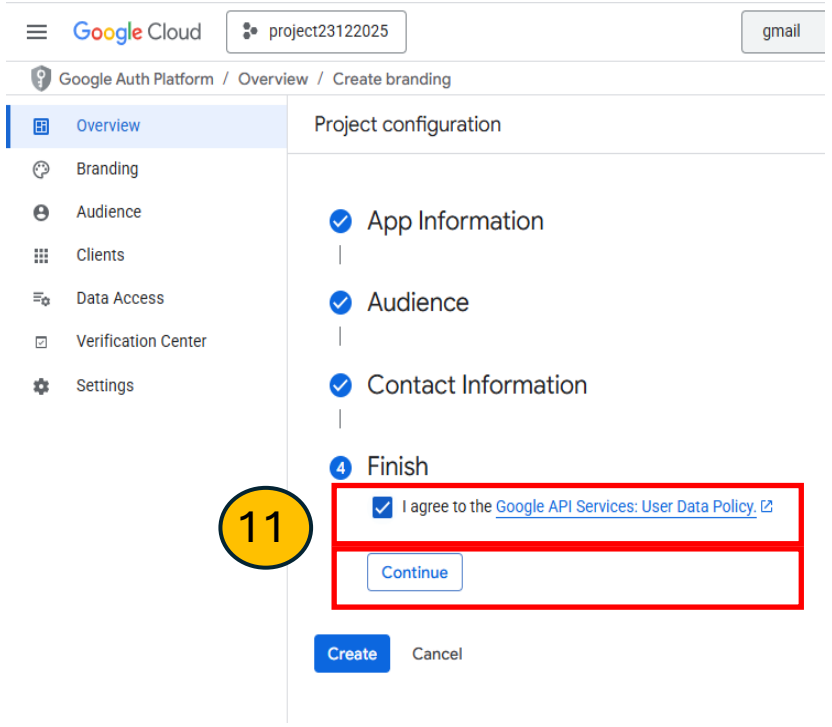
Setup Consent Screen-2/3



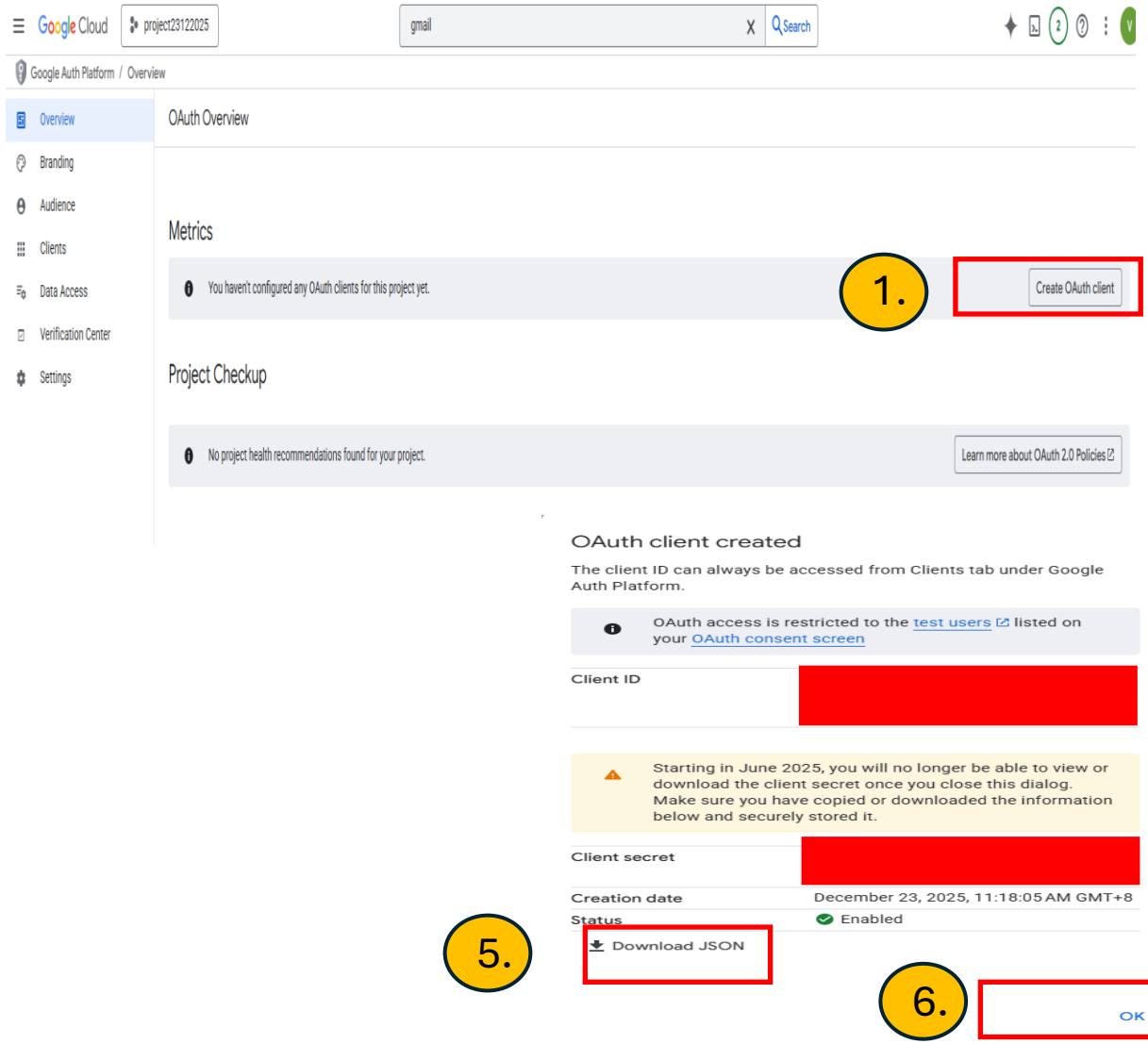


Setup Consent Screen-3/3

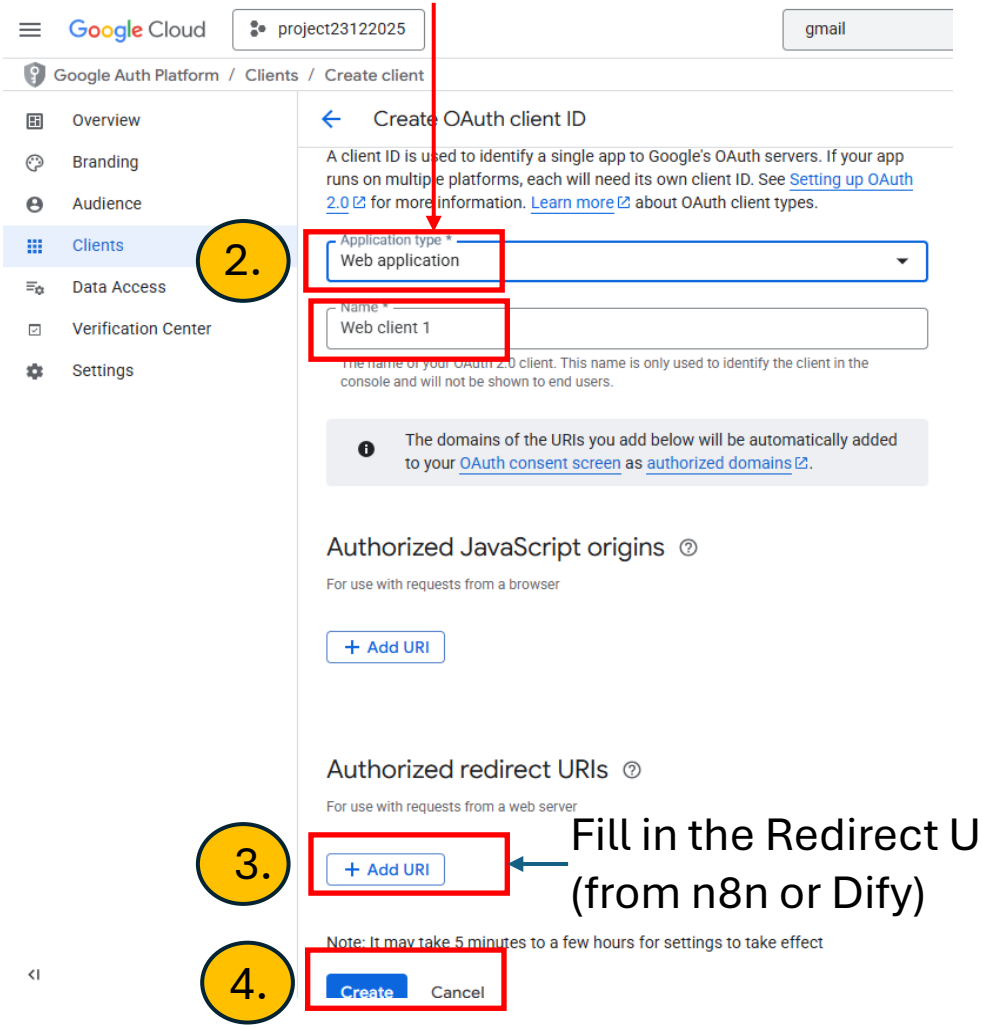




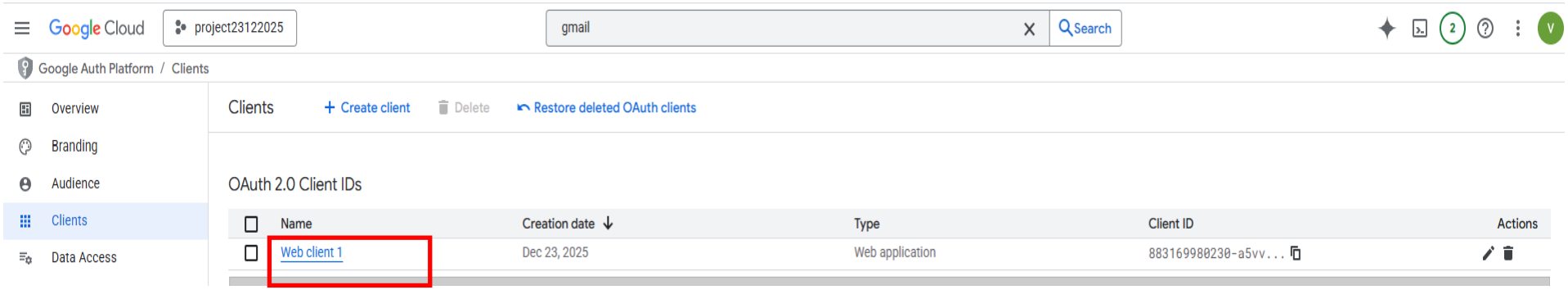
Configure OAuth Client-1/2



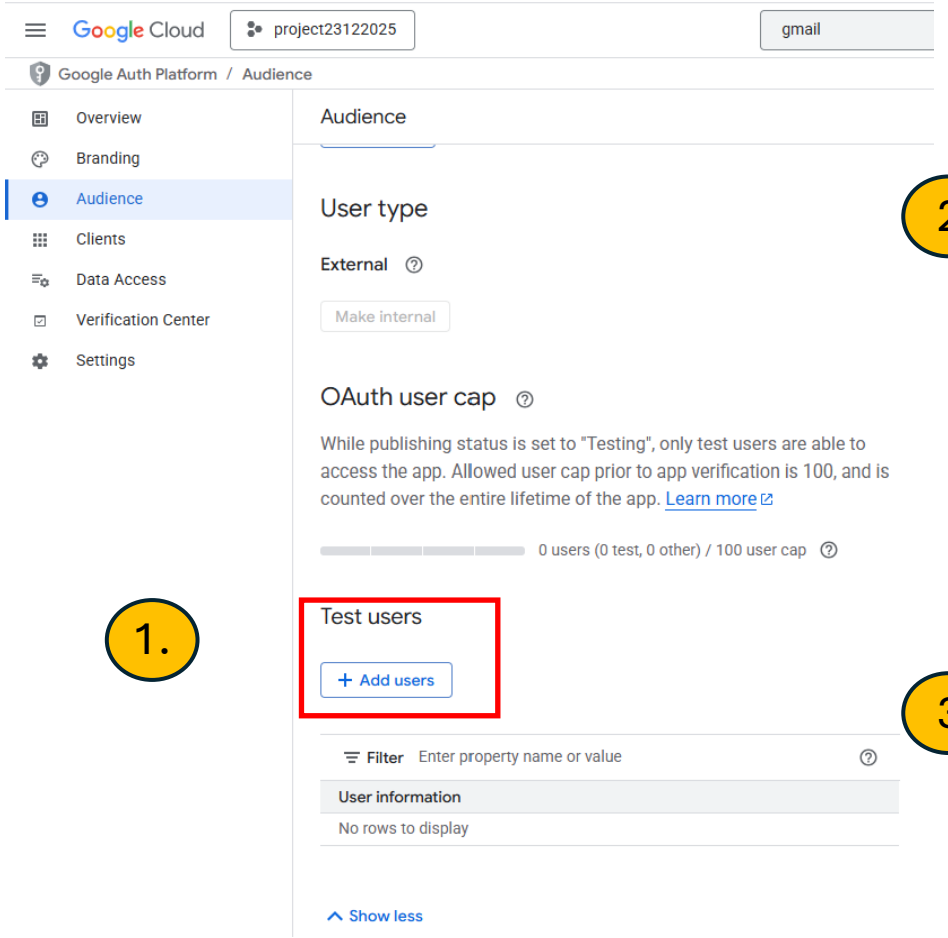
Select Web Application



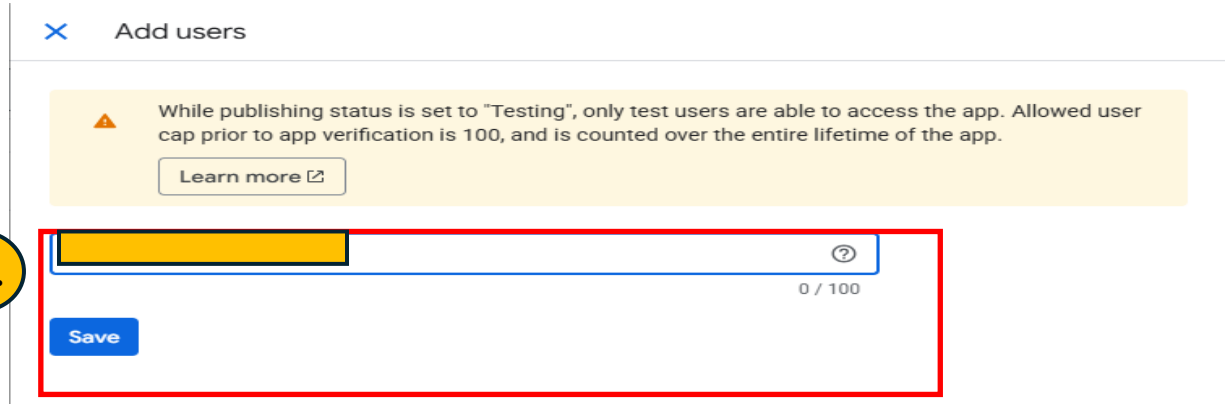
Configure OAuth Client-2/2



Create Test Users



1.



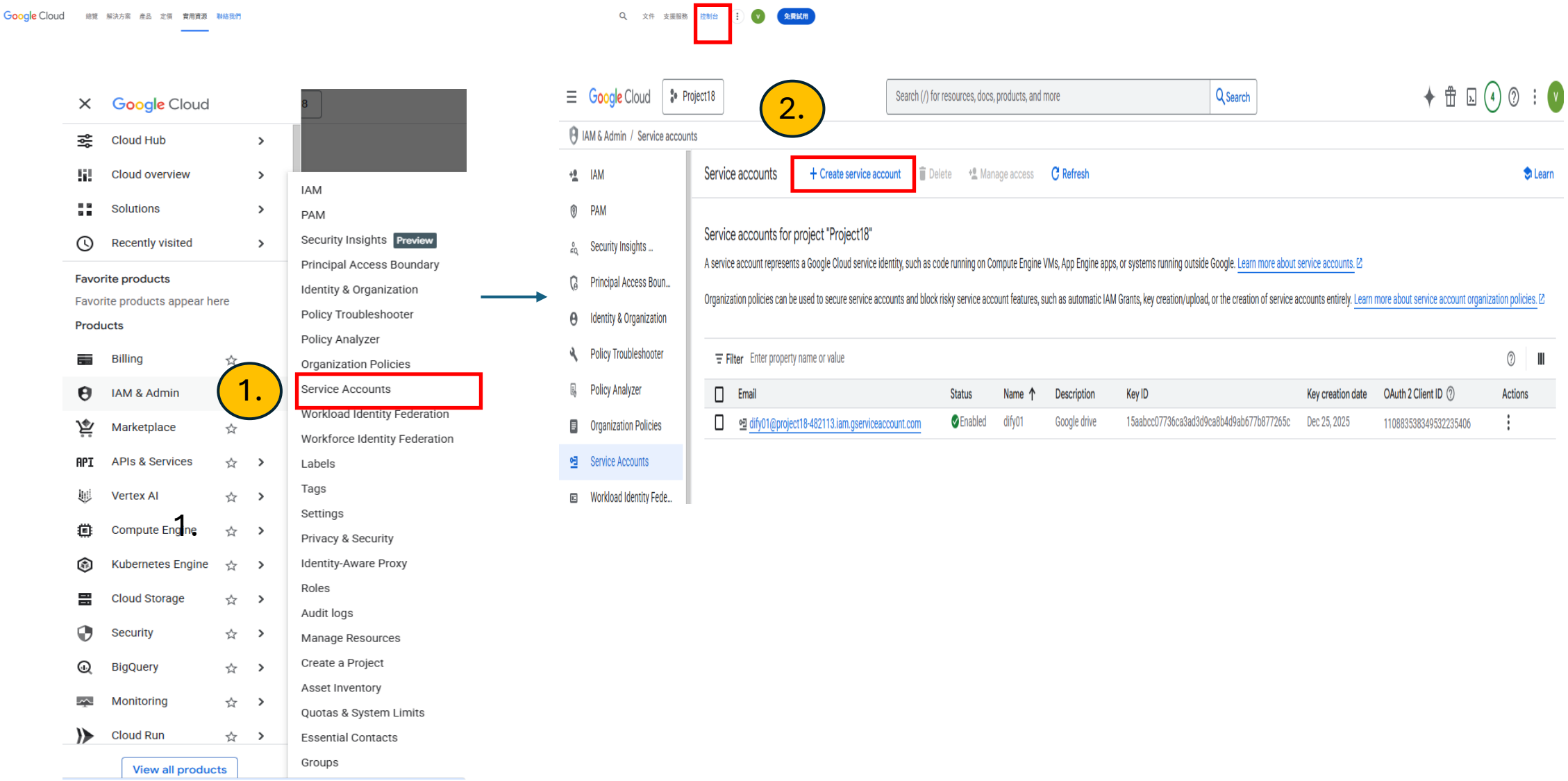
2.



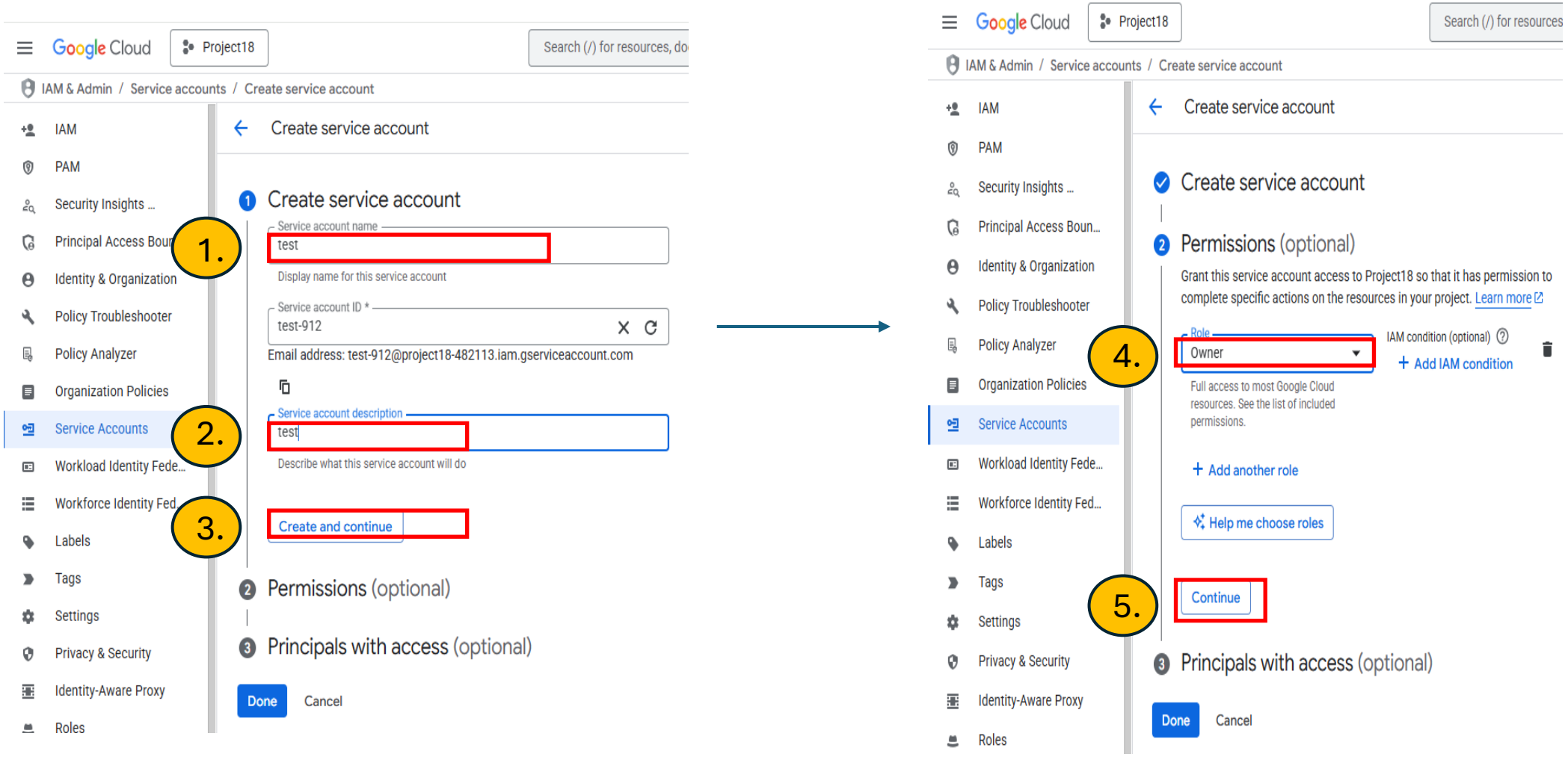
3.

Setup of Service Account

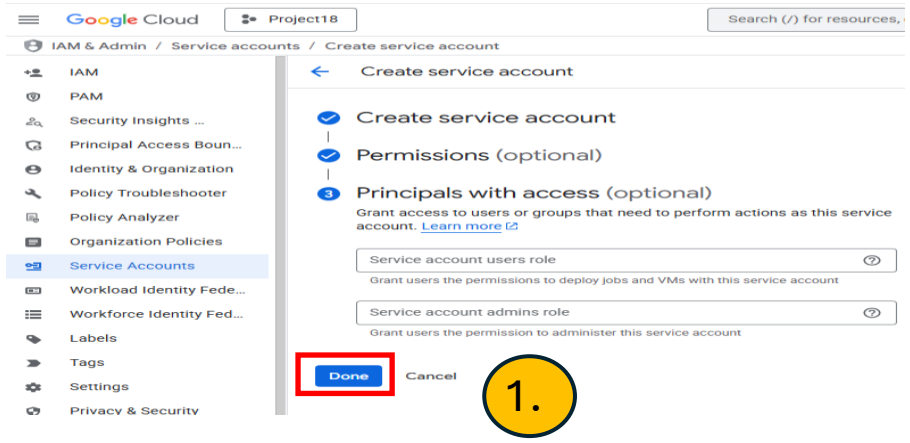
Overall Setup (Service Account)

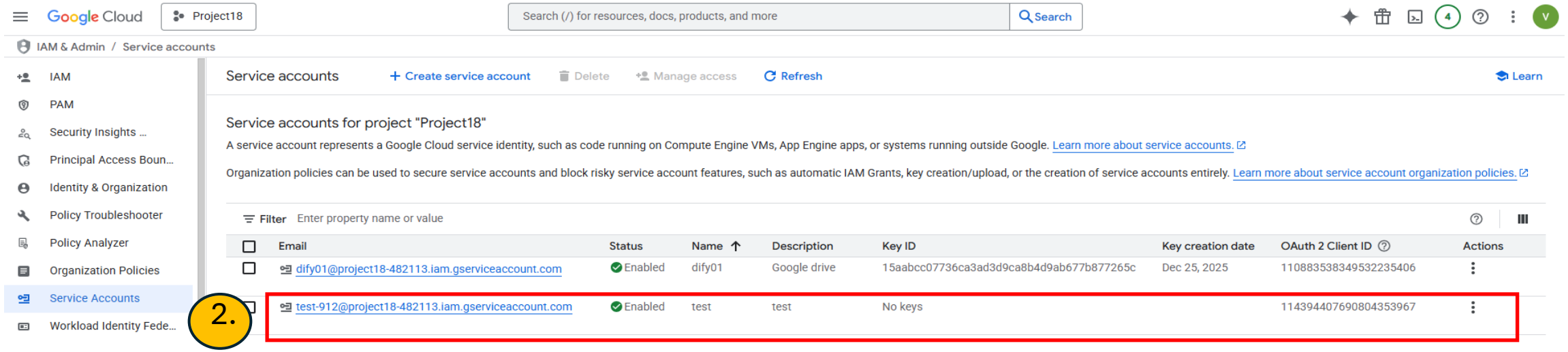


Overall Setup (Service Account)

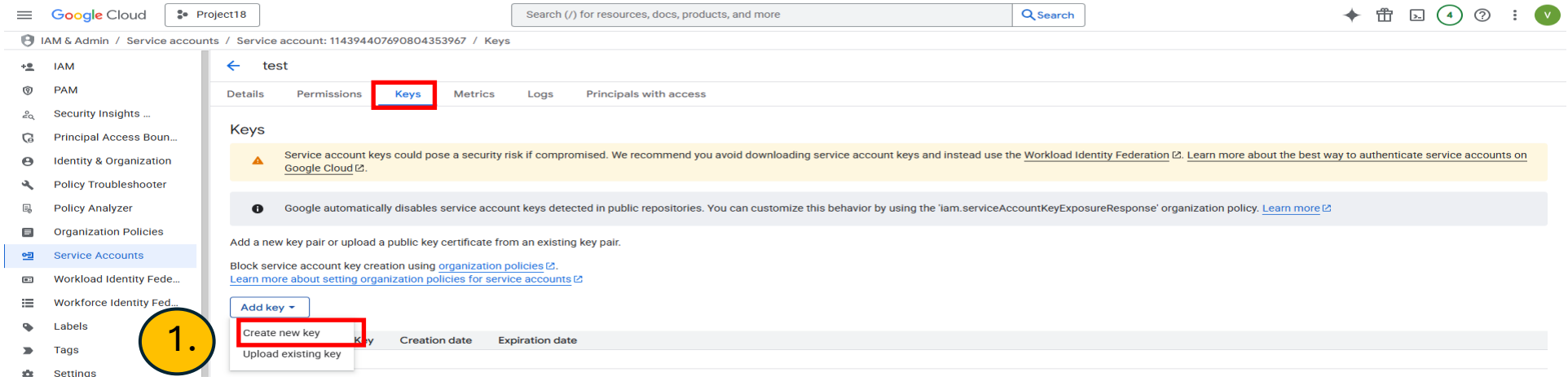


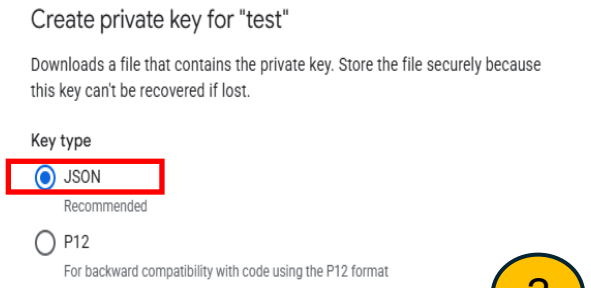
Overall Setup (Service Account)

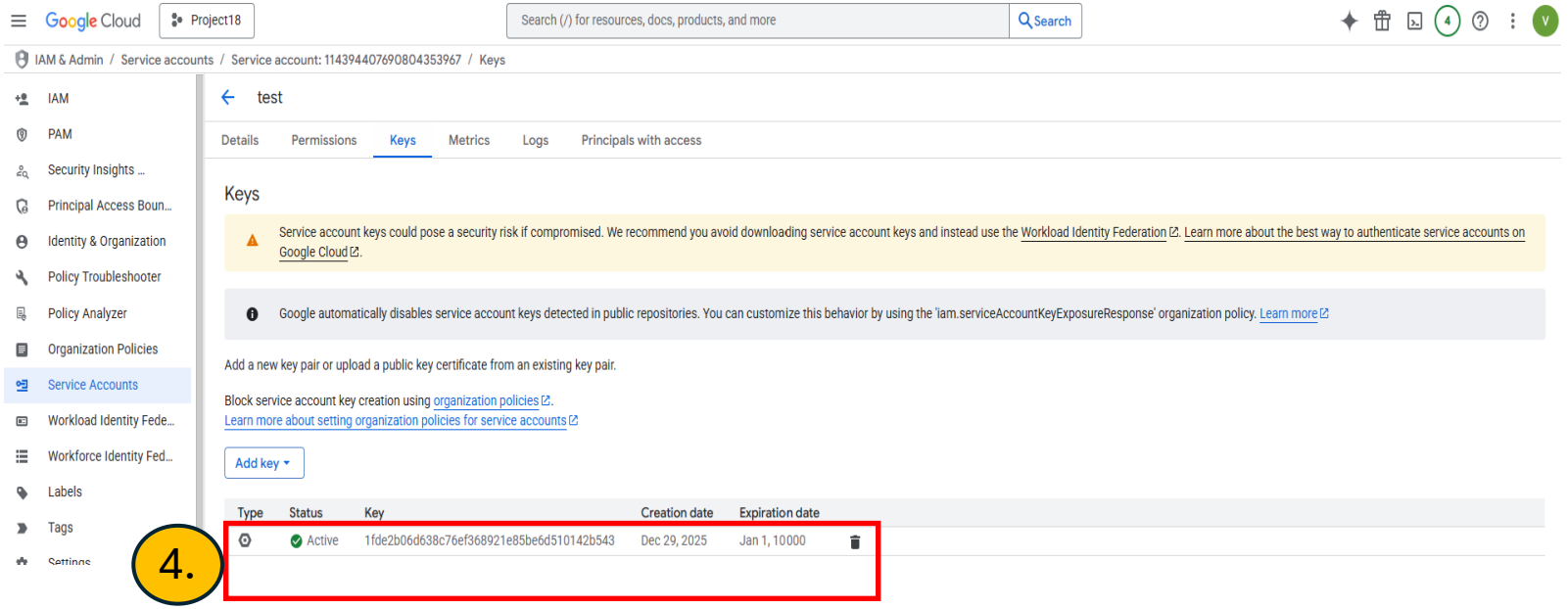




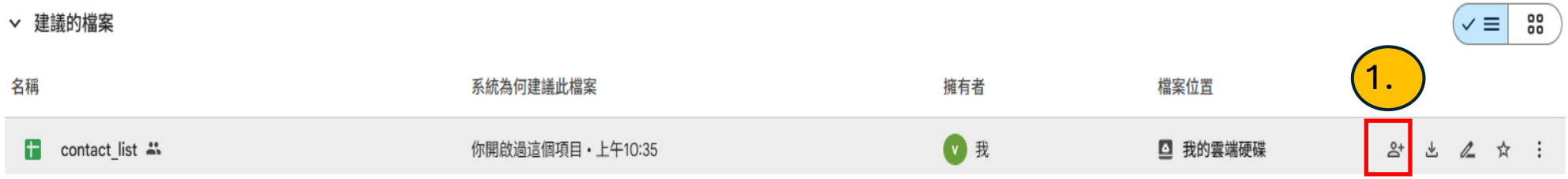
Create the key (JSON)

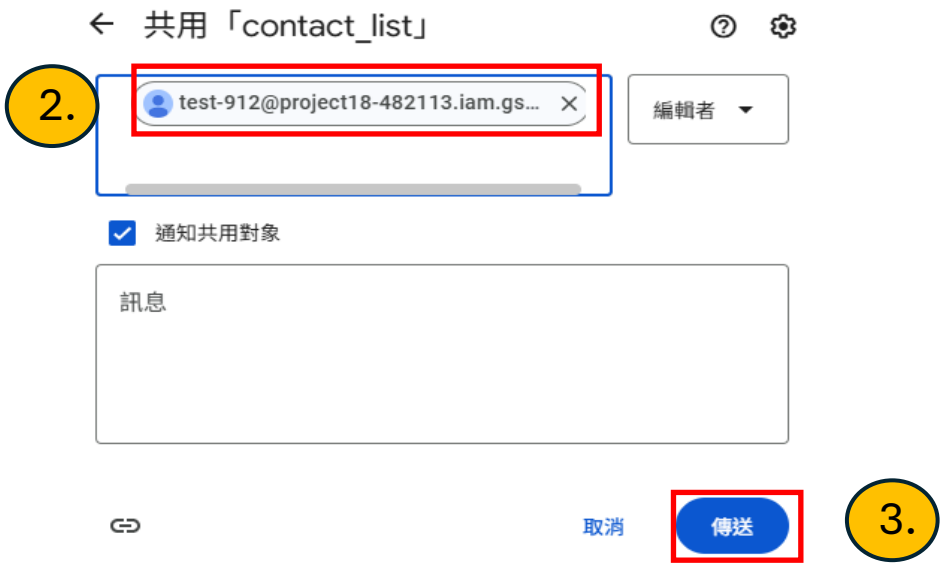




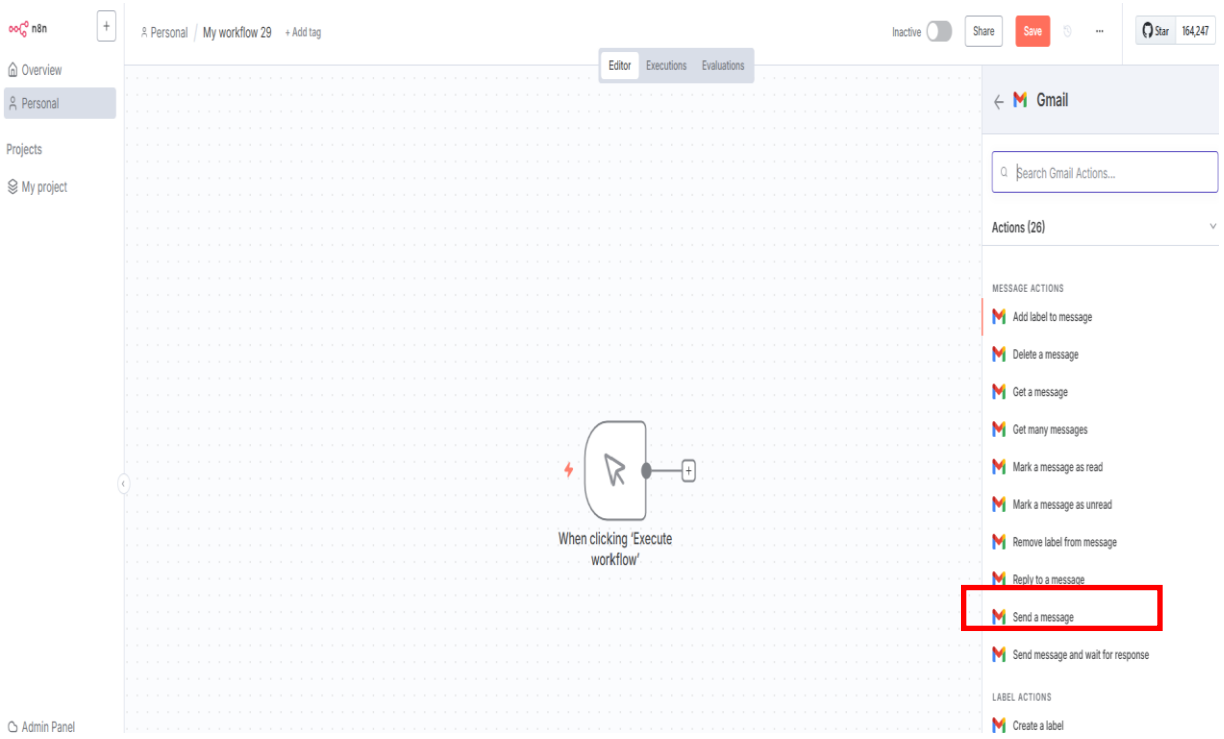


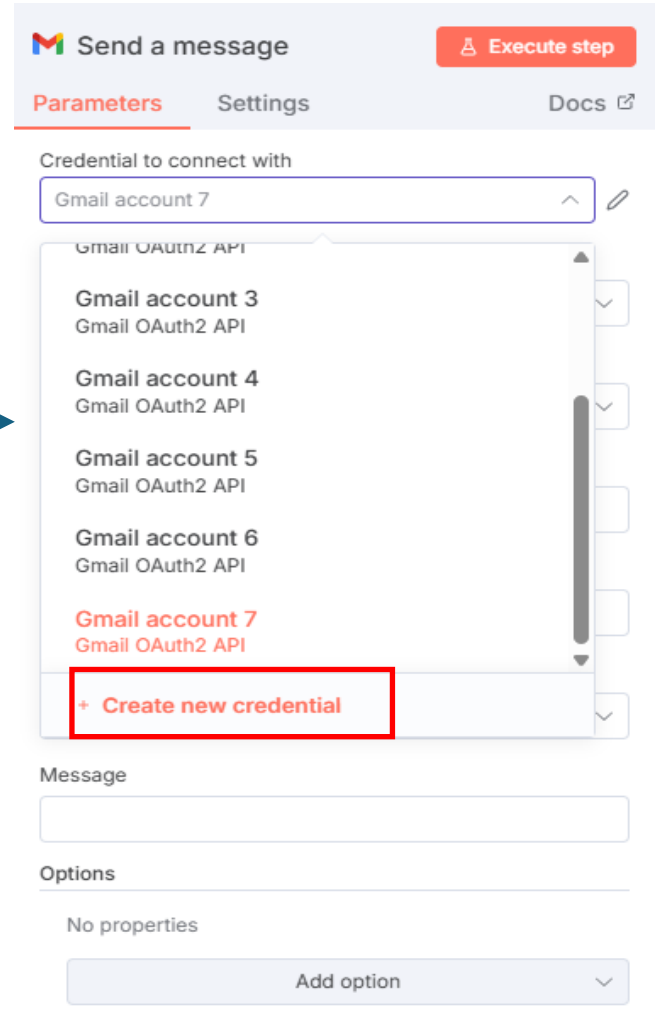
Grant Access Right to the resource (e.g. Google Sheet)



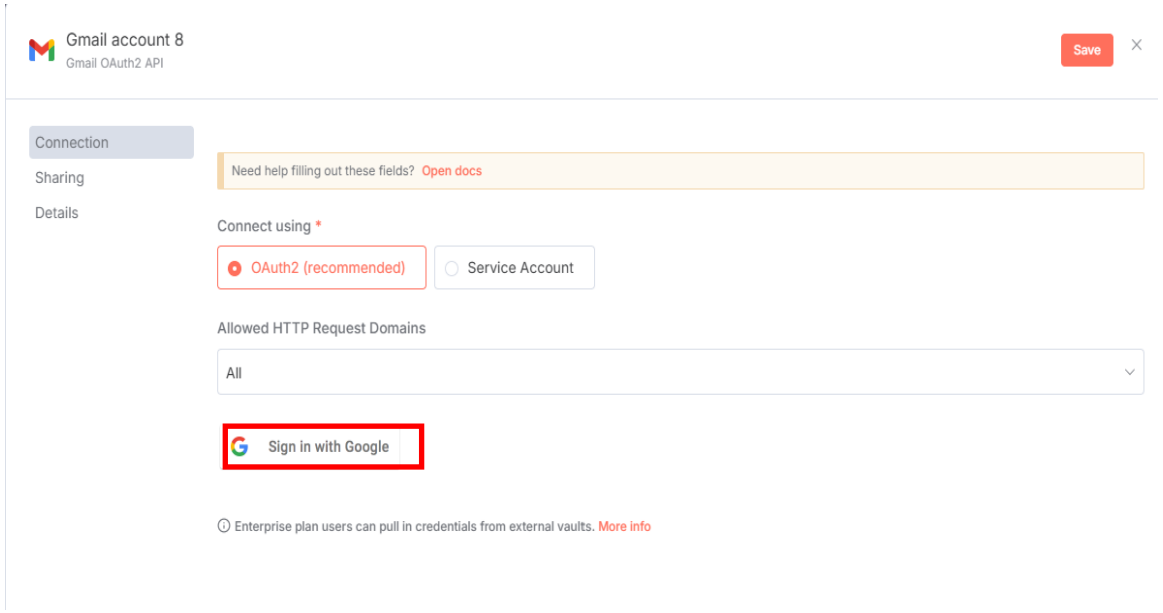


Create Gmail Service Node-1/3





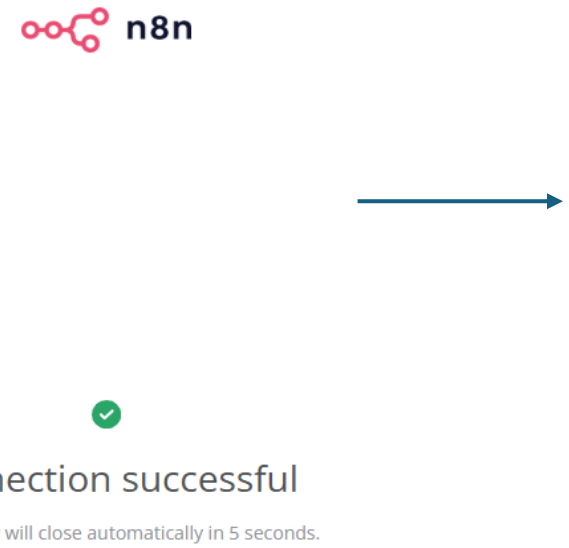
Create Gmail Service Node-2/3

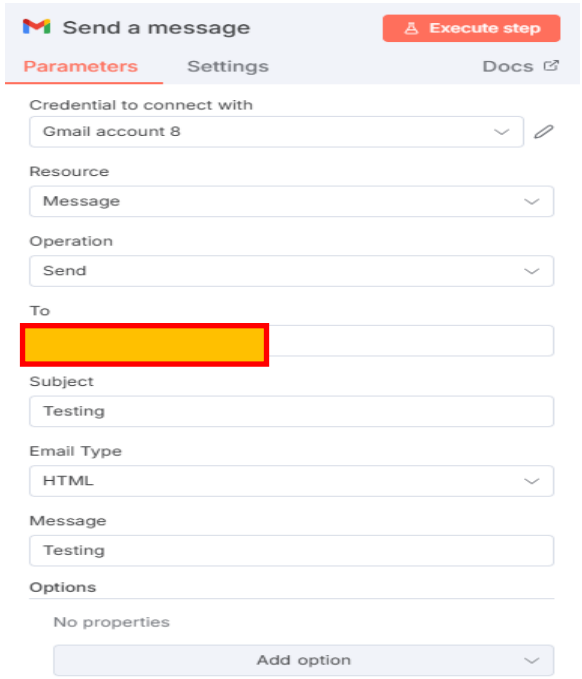




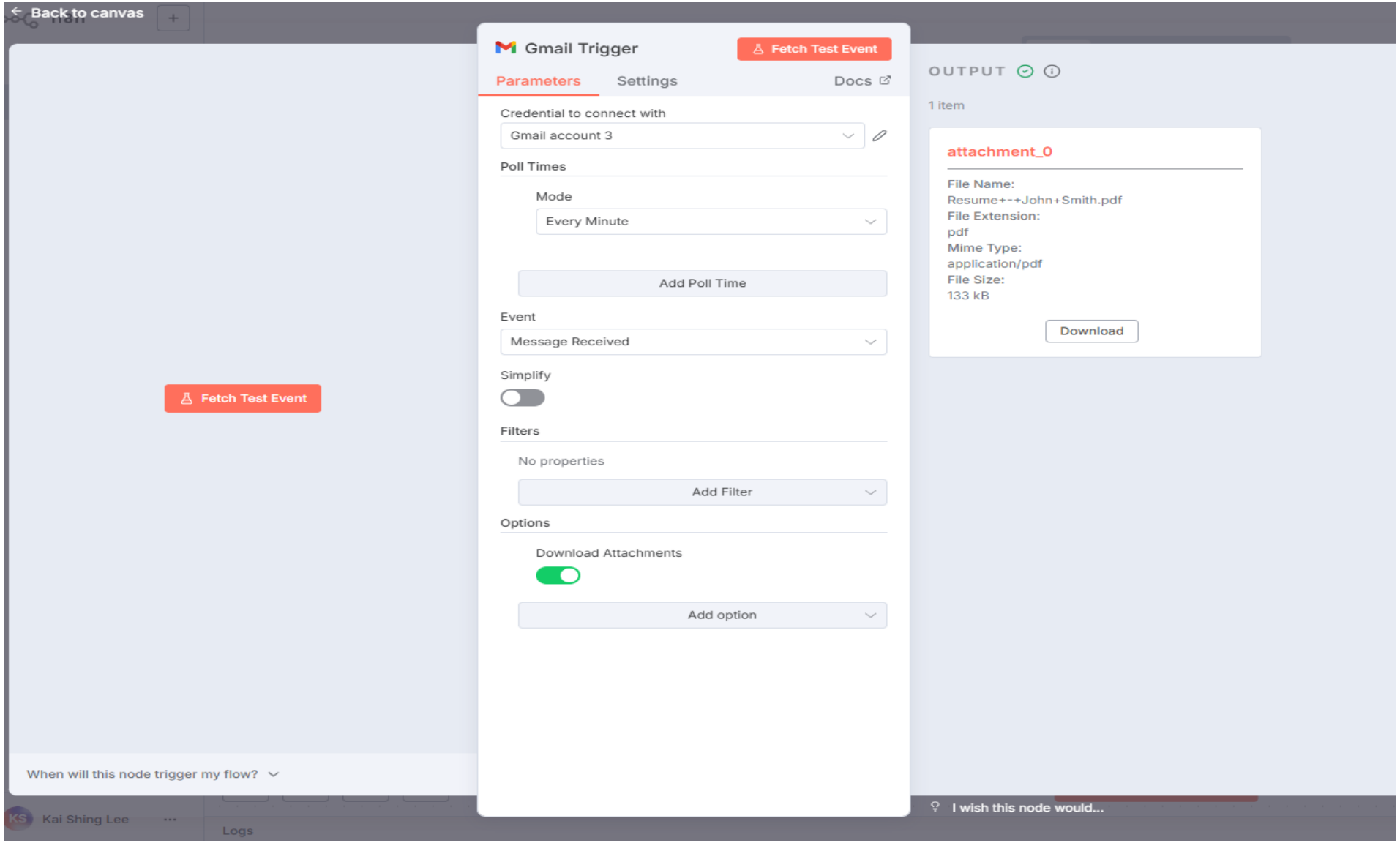
Create Gmail Service Node-3/3



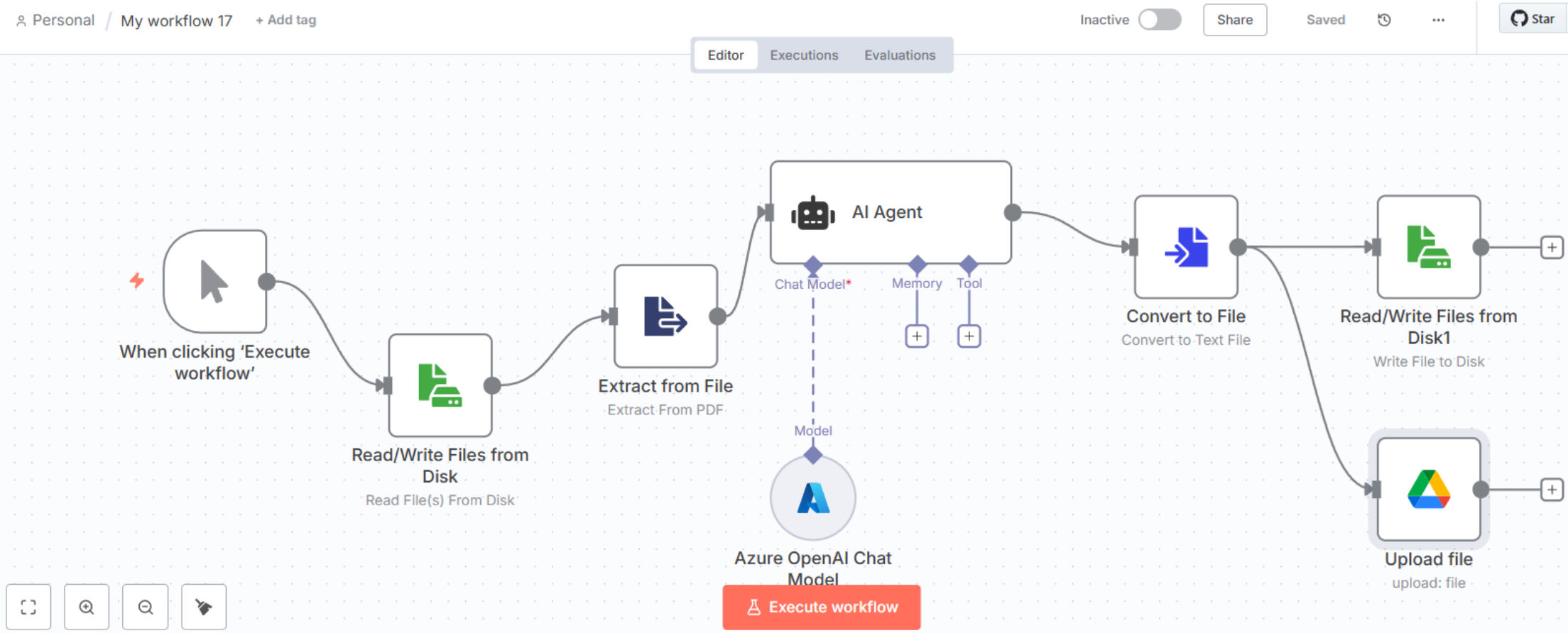




Testing of the Gmail Account



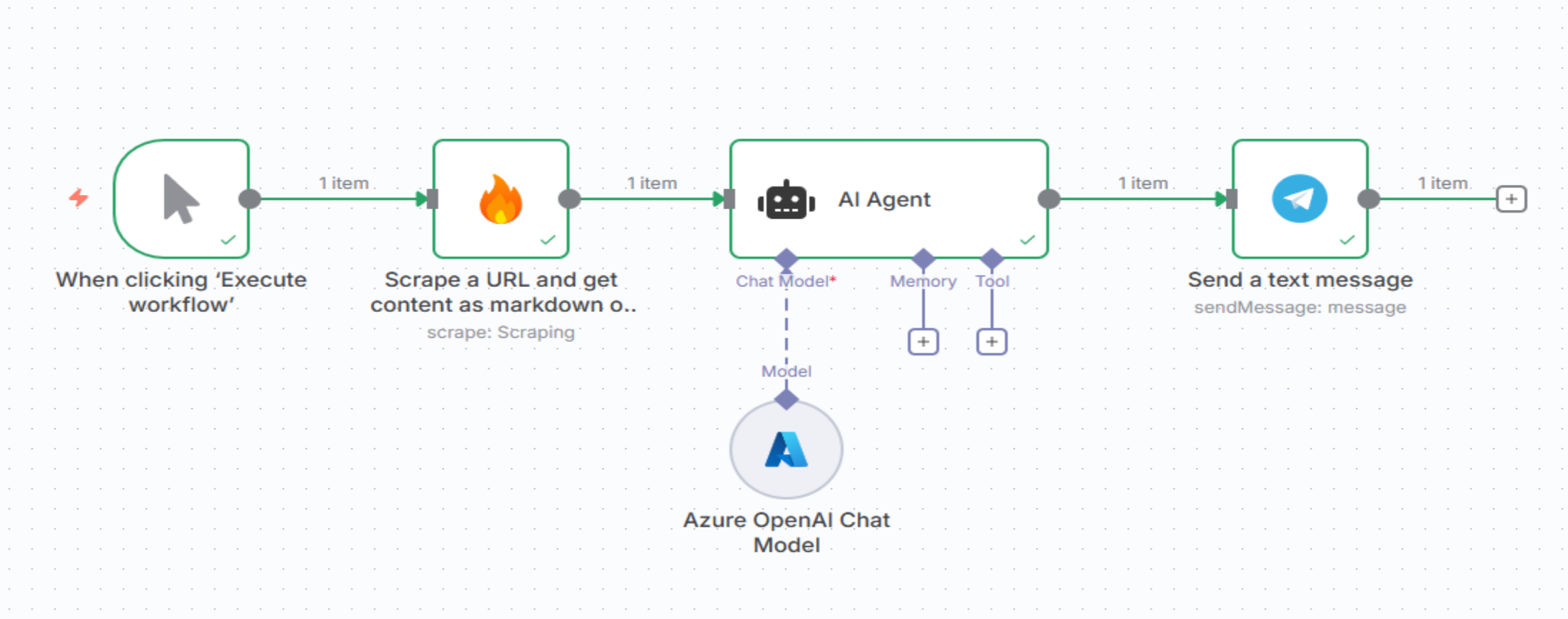
Case study
Class exercise



Web Content Summarization with LLM

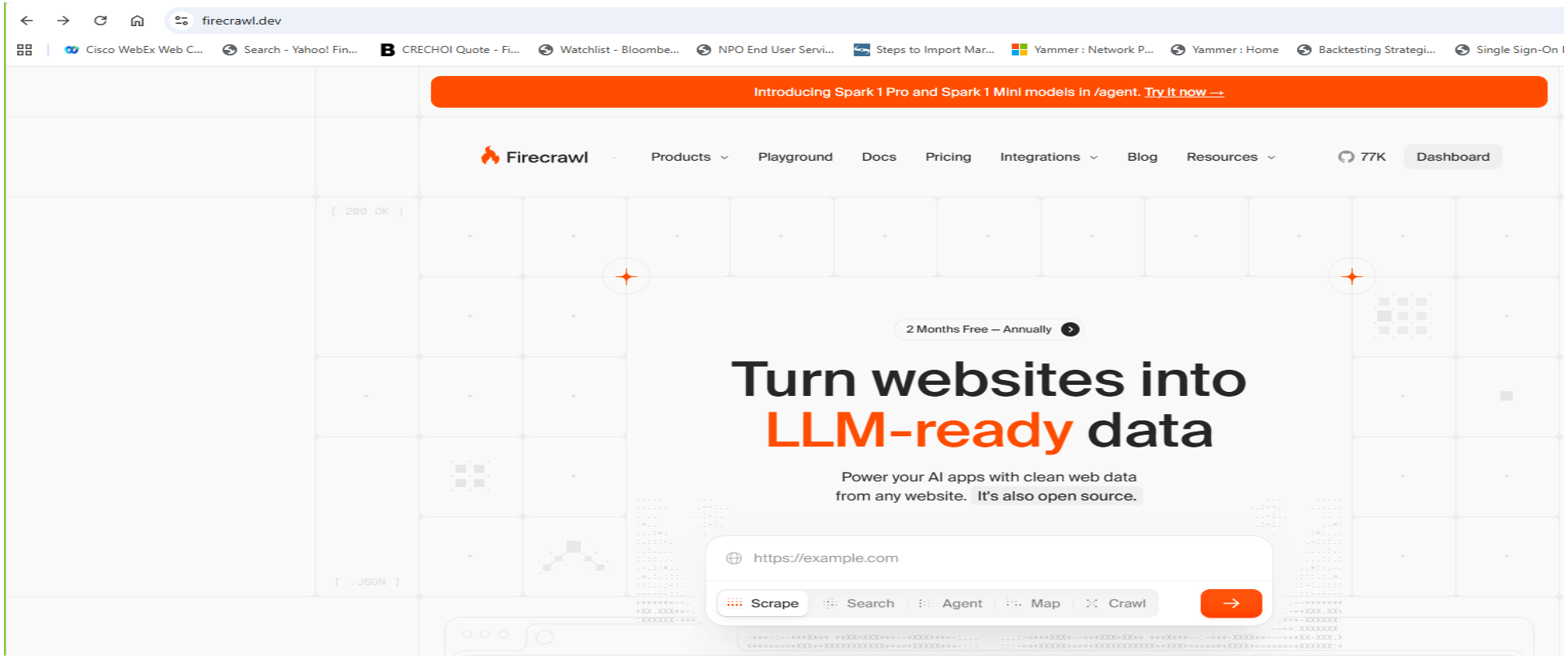
Case study

Web Content Summarization with LLM-1/3



Web Content Summarization with LLM-2/3 firecrawl

Create an account here (free API key)



Web Content Summarization with LLM-3/3

